Paired Latent Robustness Diagnostics Across Visual Stressors in World Models

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Abstract

Latent prediction is not by itself a closed-loop robustness guarantee. We develop a paired latent robustness analysis for frozen world models: clean/corrupted histories are propagated through an inspectable predictive map, summarized by a rollout-radius tail (ATR) and positive-margin proxy guard (SMPR), and related to fixed-pool planner margins. The definitions and analysis interface do not assume a particular latent-world-model architecture, but numerical calibration need not transfer across model families. A joint gate calibrated on one LeWM Gaussian sweep reaches balanced accuracy/AUPRC 0.945/0.965 on two held-out LeWM training seeds. Applied unchanged to all 24 LeWM blur/resize base-to-endpoint pairs, this absolute gate is conservative (0.733/0.856, precision 1.000, recall 0.467). For relative comparison, one untuned rule—endpoint-minus-base joint diagnostic change greater than zero—is shared across blur and resize and reaches 0.889/0.974, with Spearman correlation 0.864 between diagnostic and closed-loop robustness changes (12-block bootstrap 95% interval $[0.641, 0.932]$). The same pipeline is instantiated on one PLDM training family, but the LeWM absolute thresholds transfer weakly (0.684/0.541), supporting framework portability rather than shared calibration. A sharp candidate-wise condition covers 70.0% of 10,800 sampled fixed-pool histories. Encoder, one-step, and action/time controls remain competitive, so the evidence supports family-calibrated absolute screening and within-family cross-stressor paired comparison, not a universal absolute selector or an empirically necessary long-horizon action mechanism.

Keywords: visual world models; paired robustness diagnostics; model-family calibration; visual stressors; fixed-pool certificates.

Code and data release: `https://github.com/Anguo-star/lewm-acpc-diagnostics`

1 Introduction

JEPAs predict future representations instead of reconstructing pixels, which can reduce pressure to encode high-magnitude irrelevant visual detail [1, 3, 33]. For control, however, "irrelevant" is action- and transition-dependent. A representation may ignore a background texture but must preserve a contact pose, doorway state, or object-goal relation if that distinction changes the next useful action. Encoder-level invariance is therefore an incomplete robustness target: planning uses the model's *predictions*, so the robustness question should be asked after action-conditioned rollout.

We separate two questions that are often conflated. *Analysis-interface portability* asks whether the same paired-radius/margin construction can be instantiated for another inspectable world model. *Calibration portability* asks whether the same numerical threshold remains valid despite different latent scales, objectives, and predictor geometry. Given a clean history and a visually perturbed history for the same underlying state, we propagate both through the model's predictive interface under a shared recorded action sequence and ask whether their futures remain close enough to preserve decisions from a fixed candidate pool. The construction is architecture-agnostic under this interface, while its calibration may be model-family-specific. We use "ACPC" only as shorthand for the paired-rollout instantiation; our ablations do not establish that correct-action conditioning or a long rollout is empirically necessary for checkpoint discrimination.

The selective tension matters because consistency alone can be misleading. A collapsed model can make clean and corrupted views agree while erasing distinctions needed for planning. Conversely, a large encoder shift can be harmless if subsequent prediction contracts it before a planner cost changes. We therefore pair a same-state predictive-stability metric with a positive-margin state-proxy guard, then audit that guard with constant-collapse, label, neighbor, and action controls. Those controls support proxy-level correctness only: they do not show an incremental advantage of the task-derived labels or action sequence over their controls.

We instantiate this diagnostic with a fixed Gaussian intervention. Across PushT, TwoRoom, Reacher, and Cube, we evaluate no-noise LeWM [20] checkpoints and full-sequence input-side Gaussian-noise checkpoints over $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$. The primary endpoint is observation-only Gaussian evaluation noise with the goal image kept clean. Across three independent LeWM training seeds, the no-noise baseline remains strong on clean observations but loses 74.56 ± 5.55 points on PushT and 41.00 ± 1.44 points on Reacher at evaluation noise $\sigma = 0.08$. Test-time visual fragility is not new [16, 38, 13, 22]; our focus is what it reveals about fixed JEPA world-model checkpoints.

Within LeWM, the evidence separates the two uses. The Gaussian-frozen absolute gate transfers strongly across training seeds and conservatively across blur/resize: it makes no false passes on the 24 cross-stressor pairs but misses robust endpoints. The relative paired rule instead asks whether a checkpoint improves its diagnostic score over a reference checkpoint under the tested stressor. One zero threshold on that change is shared by blur and resize and tracks the corresponding closed-loop robustness changes across tasks and training seeds. PLDM shows that the same analysis pipeline can be executed on another latent world-model family, while its weaker absolute-gate result shows why framework portability should not be confused with shared calibration.

This paper makes three contributions. First, it defines a reproducible paired-radius/guard analysis interface and separates family-calibrated absolute screening from reference-based paired comparison; the construction applies to inspectable latent world models, while numerical thresholds are not assumed to be architecture-invariant. Second, it freezes the protocol on one LeWM Gaussian training seed and evaluates held-out LeWM seeds, all fixed-strength LeWM blur/resize pairs, one PLDM training family, and a failed repair. The cross-stressor analysis reports the single zero-threshold paired rule, continuous association, diagnostic baselines, and task–training-seed block uncertainty without stressor-specific retuning. Third, it gives candidate-wise coarse and sharp fixed-pool conditions and audits their coverage over 10,800 histories, then aligns ATR with covariance-aware JVP/finite-difference sensitivity. Encoder/H1/action/time controls and SMPR counterfactuals delimit the mechanism: they support a paired diagnostic family and fixed-pool interpretation, not empirical necessity of a long horizon, correct actions, or oracle semantics.

2 Related Work

This diagnostic sits between latent-prediction representation learning and robust visual-control methods. The related work is organized around three questions: what latent prediction promises, where augmentation-based invariance is insufficient for control, and how control-relevant representation work treats downstream predictive consequences.

2.1 Latent prediction and JEPA world models

JEPA [17] predicts in latent space rather than reconstructing pixels; I-JEPA [1] and the V-JEPA family [3, 2, 21] extend this idea to images, video, and dense physical-world representations. LeWM [20], our baseline, stabilizes an end-to-end JEPA world model with SIGReg [7]. PLDM [29, 30], via `stable-worldmodel` [19], provides related latent-dynamics planning context. LeJEPA theory [15] studies latent-variable recovery and latent planning; our diagnostic asks whether learned visual world-model checkpoints give paired clean/noised views consistent action-conditioned predictions.

2.2 Invariance, augmentation, and visual robustness

Augmentation and joint-embedding losses impose invariances only when their training signal identifies the right nuisance factors [35, 31, 40]. Recent theory clarifies why latent prediction can help with nuisance or noisy features but also why it is not enough for control. Van Assel et al. [33] and Littwin et al. [18] analyze representation-level effects; they do not address closed-loop action-conditioned rollout, candidate-cost stability, or the need to keep action-distinct states separable. Seq-JEPA [9] addresses a related invariance–equivariance tension architecturally.

We do not train robust visual-control systems as competing baselines. DrQ/DrQ-v2 [16, 38], SODA/DMC-GB [13], DreamerV3 [11], TD-MPC2 [12], and ViGMO [22] define future method-comparison axes; noisy/video

Table 1: Direct comparison with concurrent action/dynamics-consistency work. "Paired view" means a clean/corrupted same-state visual intervention; "post-hoc" means the world model is frozen. Our scope is distinguished by the paired visual stress and fixed-pool calibration, not by action consistency in general.

Work	Primary object	Paired view	Post-hoc	Changes planner	Main scope
MWM [37]	ACC/ICSD training objectives	no	no	no	navigation video generation and planning
ATM [4]	inverse-probe transfer matrix	no	yes	no	action-identifiability and within-task screening
Delta-JEPA [41]	latent-difference action decoder	no	no	no	action-sensitive, collapse-resistant training
ACID [27]	inverse-dynamics cycle residual	no	yes	yes	decision-time trajectory realizability
Future Compatible [25]	generated action–state compatibility	no	yes	selection	World Action Model rollout reliability
iKCE [26]	kinematic-null deviation	no	yes	no	physical long-horizon failure diagnosis
This work	visual paired radius and fixed-pool slack	yes	yes	no	frozen-checkpoint Gaussian stress calibration

JEPA variants [23, 14, 24] are representation-level context. ACPC is complementary: it tests whether same-state visual perturbations become equivalent after a fixed action intervention while action-, transition-, or cost-distinct cases remain discriminable.

2.3 Control-relevant predictive representations

Bisimulation merges states with equivalent reward and transition behavior. DeepMDP [8], DBC [39], bisimulation-regularized MPC [28], and Bisim-JEPA [32] use this principle to learn control-relevant representations. Value-equivalent and value-aware models [10, 34] and VIBR [6] likewise emphasize downstream control consequences. Wang et al. [36] formalize action-conditioned video world modeling as a group action with identity, inverse, and composition-consistency metrics. ReOI [5] treats visual distractor robustness in visual MPC through observation interventions for world-model action-outcome prediction. These lines show that downstream predictive consequences are already central to control-representation work. Our diagnostic instantiates that view for paired clean/noised JEPA latent-world-model rollouts: same-state noised and unperturbed views should agree after the same action sequence, while action-distinct transitions remain separable.

Our use of ACPC is therefore narrower than an identifiability theorem, a learned bisimulation metric, or a robust MPC intervention. LeJEPA-style results motivate latent prediction and planning under latent recovery assumptions; bisimulation and value-aware models define task abstractions; group-action metrics test action faithfulness; ReOI modifies observation handling for robust visual MPC. This paper instead audits additive Gaussian pixel-noise pairs in frozen checkpoints, calibrates a fixed-candidate sufficient condition, and reports when simpler representation and one-step diagnostics are equally predictive.

2.4 Concurrent action-consistency diagnostics and methods

Several concurrent papers make action or dynamics consistency central, so we do not claim priority for that general idea. MWM trains action-conditioned rollout consistency and inference-consistent diffusion distillation for image-goal navigation [37]. ATM trains fresh inverse probes on real encoded and predicted transition domains [4]; Delta-JEPA adds latent-difference action decoding during world-model training [41]; and ACID adds inverse-dynamics cycle consistency to the decision-time planning cost [27]. Ruan et al. diagnose action–state compatibility in generated World Action Model futures and identify static/background collapse as a boundary [25]. Schäfer et al. instead diagnose kinematic rather than dynamic long-horizon imagination under a physical-regime perturbation [26]. Table 1 makes the distinctions explicit.

3 Paired Robustness Diagnostics and Calibration

This section separates the portability of the analysis interface from the portability of fitted numerical thresholds, then derives the objects used in the experiments. The ideal selective criterion combines same-state perturbation agreement with different-state separation. The implemented SMPR is only a programmatic proxy for the second requirement. The fixed-pool planning link applies to the predictive-rollout instantiation; encoder and one-step quantities remain empirical component baselines rather than theorem-equivalent substitutes.

3.1 Analysis portability and two modes of use

The analysis requires paired clean/corrupted histories, an inspectable representation or predictive map, and a common diagnostic projection and normalization within each comparison. It does not require LeWM-specific layers or training losses. For a model family $\mathcal{F}$, stressor υ, and checkpoint θ, define a higher-is-better joint diagnostic score from family-calibrated radius and guard thresholds t_R, t_M:

$$S_{\mathcal{F},\upsilon}(\theta) = \min\left\{ \frac{t_R - \mathrm{ATR}_\upsilon(\theta)}{|t_R| + \varepsilon}, \frac{\mathrm{SMPR}_\upsilon(\theta) - t_M}{|t_M| + \varepsilon} \right\}. \tag{1}$$

This yields two distinct uses.

Absolute screening. The rule $S_{\mathcal{F},\upsilon}(\theta) \geq 0$ asks whether a single checkpoint passes the fitted radius and guard thresholds. Because representation scale, training objective, and predictor geometry can change across architectures, t_R, t_M are calibration parameters of $\mathcal{F}$ rather than universal constants. Holding them fixed across held-out training seeds or stressors within a family is an empirical transfer test.

Paired comparison. For an endpoint checkpoint θ_1 and a reference checkpoint θ_0 from the same family, define

$$\Delta S_{\mathcal{F},\upsilon}(\theta_1; \theta_0) = S_{\mathcal{F},\upsilon}(\theta_1) - S_{\mathcal{F},\upsilon}(\theta_0). \tag{2}$$

The natural rule $\Delta S > 0$ asks only whether the endpoint improves the paired diagnostic under stressor υ. It introduces no stressor-specific threshold, but it requires a reference checkpoint and does not certify that either checkpoint is absolutely robust. The experiments test one LeWM-calibrated score and this same zero threshold across blur and resize. PLDM is used to test whether the analysis can be instantiated on a second architecture; its one-family evidence is not used to claim shared cross-family calibration.

3.2 Paired rollout consistency and discriminability

Let o_t be an unperturbed observation and $\tilde{o}_t = \tau(o_t)$ a visually perturbed view of the *same* underlying state (τ is additive Gaussian pixel noise here, but the definitions are perturbation-agnostic). Let h_t and $\tilde{h}_t$ be the corresponding observation–action histories, E_θ the encoder, and F_θ the action-conditioned latent predictor. Fix an action sequence $\mathbf{a}_t^H = a_{t:t+H-1}$ that is *identical* on both branches. For compact notation below, $\mathbf{a}_{0:k-1}$ denotes the length-k prefix of this sequence after re-indexing time at t, and we write

$$z_t = E_\theta(h_t), \quad \tilde{z}_t = E_\theta(\tilde{h}_t), \qquad \hat{z}_{t+k} = F_\theta^k(z_t, \mathbf{a}_{0:k-1}), \quad \hat{\tilde{z}}_{t+k} = F_\theta^k(\tilde{z}_t, \mathbf{a}_{0:k-1}). \tag{3}$$

Let Π denote a *task-relevant predictive projection* — the full latent, a transition delta, goal-distance or cost features, or a learned projection — into a Euclidean diagnostic space. For nonnegative horizon weights satisfying $\sum_{k=1}^H \alpha_k = 1$, define the tuple-valued projected rollout and its canonical weighted-stacked vectorization as

$$G_{\mathbf{a}}(z) = \big(\Pi(F_\theta^1(z, \mathbf{a})), \ldots, \Pi(F_\theta^H(z, \mathbf{a}))\big), \qquad \bar{G}_{\mathbf{a}}(z) = \big[\sqrt{\alpha_1}\,\Pi(F_\theta^1(z, \mathbf{a}));\ \cdots;\ \sqrt{\alpha_H}\,\Pi(F_\theta^H(z, \mathbf{a}))\big].$$

Action-conditioned predictive consistency. For a same-state perturbation pair under the shared action sequence,

$$\mathrm{ACPC}_H(o_t, \tilde{o}_t, \mathbf{a}) = \big\|\bar{G}_{\mathbf{a}}(z_t) - \bar{G}_{\mathbf{a}}(\tilde{z}_t)\big\|_2. \tag{4}$$

For the theoretical statements below, we use the induced horizon metric

$$\begin{aligned} d_H\big((u_1, \ldots, u_H), (v_1, \ldots, v_H)\big) &= \big\| \big[\sqrt{\alpha_1}(u_1 - v_1);\ \cdots;\ \sqrt{\alpha_H}(u_H - v_H)\big] \big\|_2 \\ &= \left(\sum_{k=1}^H \alpha_k \|u_k - v_k\|_2^2 \right)^{1/2}, \qquad \alpha_k \geq 0, \quad \sum_{k=1}^H \alpha_k = 1. \end{aligned} \tag{5}$$

so that Equation (4) is exactly the clean/perturbed projected-rollout distance under the shared action sequence. In this diagnostic framing, robustness corresponds to small ACPC_H together with action-relevant discriminability, not to the encoder distance $d(z_t, \tilde{z}_t)$ alone. Encoder geometry remains an indispensable first-stage risk signal: noisy views should stay in the same-state predictive basin and avoid crossing into task-distinct neighborhoods. The narrower point is that raw encoder distance alone is not a complete robustness criterion, because the effect of an encoder shift depends on local action-conditioned rollout sensitivity and candidate margins.

Action-relevant discriminability. Consistency must be *selective*. Let $Y^H(s, \mathbf{a})$ denote a task-relevant future signal that is external to the learned predictor — simulator state, cost-to-go, inverse-dynamics label, contact/keyframe tag, or another diagnostic proxy available in the dataset. For two genuinely different states s_i, s_j (with latents z_i, z_j) that admit an action sequence under which this external future signal diverges,

$$\Delta^H_{\mathrm{dyn}}(s_i, s_j, \mathbf{a}) = d_Y\big(Y^H(s_i, \mathbf{a}),\, Y^H(s_j, \mathbf{a})\big) > m, \tag{6}$$

the representation and predictions should keep them separable,

$$d\big(\Psi(z_i),\, \Psi(z_j)\big) > m', \tag{7}$$

where Ψ is a discriminability projection, possibly action-conditioned (latent, predicted delta under $\mathbf{a}$, inverse-dynamics feature, cost feature, or rollout embedding), and $m, m' > 0$ are margins. Equations (4)–(7) state the target: lower predictive disagreement for same-state visual perturbations must preserve discriminability among state/action/transition-distinct pairs. A method that drove $\mathrm{ACPC}_H \to 0$ by collapsing the latent would violate (7), so the two conditions are inseparable.

The ideal guard can also be posed over state–action pairs: if two pairs have external task-relevant futures that differ by more than a margin, their projected rollouts should remain separated. The reported SMPR does not validate this action-faithfulness condition: it uses programmatic label-crossing state pairs, and action permutation produces no SMPR change on the MVE.

3.3 Predictive tubes and planner margins

The selective condition can be written as a radius–margin event. For anchor i, let $u^{\mathrm{obs}}_{i,0}$ be the projected representation of the last clean history observation and let $u^{\mathrm{obs}}_{i,1:H}$ be those of the clean observed future. The per-anchor clean transition scale is

$$s_i = Q_{0.50}\left(\left\{\left\|u^{\mathrm{obs}}_{i,k} - u^{\mathrm{obs}}_{i,k-1}\right\|_2 : k = 1, \ldots, H\right\}\right), \tag{8}$$

so the transition from the last history observation to the first clean future observation is included. For noise draw m at perturbation level σ, the canonical normalized horizon radius is

$$\begin{aligned} R^{(i,m)}_\sigma &= \frac{1}{s_i + \varepsilon}\big\|\bar{G}_{\mathbf{a}_i}(E_\theta(h_i)) \\ &\qquad\qquad - \bar{G}_{\mathbf{a}_i}(E_\theta(\tilde{h}^{(m)}_{i,\sigma}))\big\|_2, \\ R^{(i)}_\sigma &= \frac{1}{M}\sum_{m=1}^{M} R^{(i,m)}_\sigma, \qquad \mathrm{ATR}_q(\sigma) = r_q(\sigma) = Q_q\left(\left\{R^{(i)}_\sigma\right\}_{i=1}^{n}\right). \end{aligned} \tag{9}$$

Thus multiple draws are aggregated conditionally within each anchor before the checkpoint-level quantile. The radius side asks whether a visual perturbation remains inside the same-state predictive tube under the recorded action sequence. The margin side asks whether planner and state-proxy distinctions remain outside that tube. The cost-Lipschitz statements below use the corresponding unnormalized distance $D_{i,m} = (s_i + \varepsilon)R^{(i,m)}_\sigma$, so its units match the planner cost. For a fixed candidate pool $\mathcal{A}$, the clean planner margin is

$$\Delta_{\mathcal{A}}(h, g) = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g), \tag{10}$$

where lower cost is preferred and $\mathbf{a}^{(1)}, \mathbf{a}^{(2)}$ are the clean top-1 and top-2 candidates. For programmatically selected different-state pairs, we write the proxy margin as

$$M_{\mathrm{diff}}(i, j, \mathbf{a}) = d(\Psi(z_i), \Psi(z_j))\,, \tag{11}$$

and track the failure event $M_{\text{diff}} \leq R_\sigma + \delta_m$ for a reporting margin $\delta_m \geq 0$; when R_σ is normalized, the different-state distance and δ_m use the same anchor scale. The public-v1 protocol fixes $\delta_m = 0.10$. Small radius without margin is compatible with collapse, while large proxy margin without small radius is visually unstable planning. The proxy controls later prevent either side from being read as semantic sufficiency.

3.4 Fixed-pool radius–margin link to planning

ACPC is connected to planning through the candidate costs evaluated by model-predictive control. Let h be a clean history, $\tilde{h}$ the corresponding perturbed history, and let both branches evaluate the same candidate action sequence $\mathbf{a}$. For a goal or task descriptor g, define

$$C_h(\mathbf{a}, g) = J(\Pi(\hat{z}_{t+1}), \ldots, \Pi(\hat{z}_{t+H}), g)\,, \tag{12}$$

where J is the planner cost on the projected-rollout sequence. $C_{\tilde{h}}(\mathbf{a}, g)$ is defined analogously from the perturbed branch. The following statements are sufficient conditions for stability on a fixed candidate set; they do not prove stability of the full CEM sampling distribution or of the closed-loop trajectory over repeated replanning.

Encoder geometry enters the planning link through the composed rollout map

$$G_{\mathbf{a}}(z) = \Pi(F_\theta^{1:H}(z, \mathbf{a})).$$

Smaller same-state encoder shifts and reduced neighborhood crossing lower upstream risk, but the planning-relevant quantity is the projected-rollout distance after the shift is passed through $G_{\mathbf{a}}$. The empirical diagnostic therefore reports action-conditioned rollout-disagreement tails rather than raw encoder distances alone.

If the planner cost J is locally L_J-Lipschitz on a fixed clean/perturbed projected-rollout neighborhood, then a paired rollout discrepancy at most ϵ implies candidate-cost drift at most $L_J\epsilon$:

$$d_H\big(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a})), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}))\big) \leq \epsilon \quad \Rightarrow \quad |C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \leq L_J\epsilon.$$

For a shared candidate set, if every candidate cost moves by at most η and the clean top-1/top-2 margin Δ is larger than 2η, the top candidate is unchanged. Equivalently, the ACPC-cost condition gives fixed-candidate stability when $\Delta > 2L_J\epsilon$. These are fixed-candidate statements only; adaptive sampling, repeated replanning, and environment feedback can still change the closed-loop trajectory. Formal statements and proofs are in Appendix A.

The sample-level audit also uses a sharper candidate-wise condition. Let j^* be the clean top candidate, $\Delta_j = C_h(\mathbf{a}^j, g) - C_h(\mathbf{a}^{j^*}, g)$, and $d_j = |C_{\tilde{h}}(\mathbf{a}^j, g) - C_h(\mathbf{a}^j, g)|$. If

$$\min_{j \neq j^*}\big(\Delta_j - d_j - d_{j^*}\big) > 0, \tag{13}$$

then every competitor remains more costly than j^* on the perturbed branch. This implication is exact for the evaluated fixed pool and deterministic tie-breaking. Its empirical coverage is not a probability theorem for a newly sampled or adaptive pool.

Theorem 1 (Fixed-pool radius–margin bound)**.** *Let $\mathcal{A} = \{\mathbf{a}^1, \ldots, \mathbf{a}^K\} \sim q^K$ be a once-sampled candidate pool with deterministic tie-breaking. For a clean history h and matched perturbed history $\tilde{h}$, define*

$$D_j = d_H\big(G_{\mathbf{a}^j}(E_\theta(h)), G_{\mathbf{a}^j}(E_\theta(\tilde{h}))\big)\,.$$

Assume the planner cost J is locally L_J-Lipschitz on the evaluated projected-rollout neighborhood and that for a single sampled candidate

$$\Pr_{\mathbf{a}\sim q}[D(\mathbf{a}) > \epsilon] \leq \alpha.$$

Let $\Delta_{\mathcal{A}} = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g)$ be the clean top-1/top-2 candidate margin. Then

$$\Pr_{\mathcal{A}}\left[\arg\min_j C_h(\mathbf{a}^j, g) \neq \arg\min_j C_{\tilde{h}}(\mathbf{a}^j, g)\right] \leq K\alpha + \Pr_{\mathcal{A}}[\Delta_{\mathcal{A}} \leq 2L_J\epsilon].$$

If the state-proxy discriminability failure event satisfies

$$\Pr[M_{\text{diff}} \leq R_\sigma + \delta_m] \leq \gamma,$$

then the selective-ACPC diagnostic failure risk is controlled by

$$\Pr[\text{selective failure}] \leq K\alpha + \Pr_{\mathcal{A}}[\Delta_{\mathcal{A}} \leq 2L_J\epsilon] + \gamma.$$

This is a fixed-pool diagnostic bound for a matched perturbation distribution. It places a checkpoint in the bound's sufficient-condition regime only when the high-probability same-state predictive radius is below the relevant planner margin and the measured proxy separations remain outside the perturbation tube.

Definition 1 (Matched-perturbation diagnostic region)**.** *For a checkpoint family indexed by training perturbation level ρ and evaluated at observation noise σ, define the empirical radius–margin pass event*

$$\widehat{\mathcal{G}}_{\rho,\sigma} = \left\{\widehat{Q}_{1-\alpha}(R_\sigma) \le \widehat{r}_{\mathrm{margin}}\right\} \cap \left\{\widehat{\mathrm{SMPR}}_{\rho,\sigma} \ge 1 - \widehat{\gamma}\right\}.$$

The matched-perturbation diagnostic region is the set of checkpoint indices ρ for which $\widehat{\mathcal{G}}_{\rho,\sigma}$ holds. If planner-margin traces are unavailable, $\widehat{r}_{\mathrm{margin}}$ must be reported as a rollout-space proxy, not as a calibrated planner-margin bound. If diagnostics are available only at the no-noise and recovered endpoints, the evidence is endpoint mechanism evidence rather than interval identification.

3.5 Sampled-pool tails and local Gaussian sensitivity

The fixed-candidate statements motivate the deterministic link from paired rollout agreement to candidate decisions. A once-sampled candidate pool adds a tail requirement: if a single candidate has $\Pr[D > \epsilon] \le \delta$, then a pool of size K can still contain a bad clean/noisy rollout pair with probability up to $K\delta$. A flip can also occur when the clean top-1/top-2 margin is at most $2L_J\epsilon$. This is the reason ATR uses an upper-tail summary rather than a mean disagreement score, and it is also why ATR is not a closed-loop control guarantee. Appendix A gives the proof.

The Gaussian stressor also has a local sensitivity interpretation. We use the same weighted-stacked map $\bar{G}_{\mathbf{a}}$ defined above and let $\tilde{o} = o + \xi$ with $\xi \sim \mathcal{N}(0, \sigma^2 I)$ in raw pixel coordinates. Here E_θ includes the deterministic ImageNet input normalization; equivalently, JVP tangents in model-input coordinates are scaled by the reciprocal per-channel ImageNet standard deviation.

Proposition 1 (Local Gaussian ACPC sensitivity)**.** *If E_θ and $\bar{G}_{\mathbf{a}}$ are locally twice differentiable around o and $E_\theta(o)$ with bounded second-order remainders, then for small σ,*

$$\bar{G}_{\mathbf{a}}(E_\theta(o+\xi)) - \bar{G}_{\mathbf{a}}(E_\theta(o)) = J_{\bar{G}_{\mathbf{a}}}(E_\theta(o)) J_E(o)\xi + R_\xi,$$

where the remainder contributes higher-order terms, and

$$\mathbb{E}_\xi\left[\left\|\bar{G}_{\mathbf{a}}(E_\theta(o+\xi)) - \bar{G}_{\mathbf{a}}(E_\theta(o))\right\|_2^2\right] = \sigma^2 \left\|J_{\bar{G}_{\mathbf{a}}}(E_\theta(o)) J_E(o)\right\|_F^2 + O(\sigma^3).$$

The same linearization gives the local tail interpretation used by ATR.

Corollary 1 (Local Gaussian ACPC radius quantile)**.** *Under the local linearization in Proposition 1, condition on an anchor history h, its clean transition scale $s(h)$, and the shared action sequence, and let*

$$A_{h,\mathbf{a}} = J_{\bar{G}_{\mathbf{a}}}(E_\theta(o)) J_E(o),$$

and let $\lambda_1, \ldots, \lambda_r$ be the nonzero eigenvalues of $A_{h,\mathbf{a}} A_{h,\mathbf{a}}^\top$. For isotropic pixel noise $\xi \sim \mathcal{N}(0, \sigma^2 I)$,

$$R_\sigma^2(h, \mathbf{a}) \approx \frac{\sigma^2}{(s(h) + \varepsilon)^2} \sum_{i=1}^{r} \lambda_i \chi_i^2,$$

and therefore

$$r_{1-\alpha}(\sigma; h, \mathbf{a}) = Q_{1-\alpha}(R_\sigma) \approx \frac{\sigma}{s(h) + \varepsilon} \sqrt{Q_{1-\alpha}\left(\sum_i \lambda_i \chi_i^2\right)}.$$

This corollary connects the empirical ATR quantile to the singular spectrum of the action-conditioned encoder–predictor Jacobian. ATR decreases when the composed map contracts along nuisance-noise directions, not necessarily when raw encoder distance alone is minimized. A useful diagnostic decomposition is

$$S_{\mathrm{comp}}(h, \mathbf{a}) = \left\|J_{\bar{G}_{\mathbf{a}}}(E_\theta(o)) J_E(o)\right\|_F^2 = \mathrm{tr}\left(J_E^\top J_{\bar{G}_{\mathbf{a}}}^\top J_{\bar{G}_{\mathbf{a}}} J_E\right),$$

with encoder-side and rollout-side reference scales

$$S_E(o) = \|J_E(o)\|_F^2, \qquad S_{\bar{G}}(E_\theta(o), \mathbf{a}) = \|J_{\bar{G}_\mathbf{a}}(E_\theta(o))\|_F^2.$$

The main mechanism analysis prioritizes the raw encoder, rollout, and composed traces S_E, $S_{\bar{G}}$, and S_{comp}. For a scale-relative appendix diagnostic, define both

$$\kappa_{\mathrm{sub}}(h, \mathbf{a}) = \frac{\|J_{\bar{G}_\mathbf{a}} J_E\|_F^2}{\|J_{\bar{G}_\mathbf{a}}\|_F^2 \|J_E\|_F^2 + \varepsilon} \le 1, \qquad \kappa_{\text{relative isotropic}}(h, \mathbf{a}) = d_z \, \kappa_{\mathrm{sub}}(h, \mathbf{a}),$$

where the first inequality follows for exact maps from Frobenius-norm submultiplicativity, and d_z is the full intermediate latent input dimension used by these Jacobian maps. The relative isotropic gain can exceed one; neither ratio supersedes the raw-trace mechanism evidence.

Combining the local radius quantile with the fixed-pool margin condition gives the diagnostic noise scale

$$\sigma^*(h, \mathbf{a}, \mathcal{A}) = \frac{\Delta_\mathcal{A}}{2L_J \sqrt{Q_{1-\alpha}(\sum_i \lambda_i \chi_i^2)}}.$$

This is a local diagnostic approximation, not a global robustness theorem. When L_J or the weighted-chi-square tail is unavailable, the empirical cost-drift slope $\widehat{s}_C = \widehat{Q}_q(|C_h - C_{\tilde{h}}|)/\sigma$ gives only a proxy tolerance $\widehat{\sigma}^*_{\mathrm{cost}} = \widehat{Q}_{0.50}(\Delta_\mathcal{A})/(2\widehat{s}_C)$.

Thus Gaussian ACPC measures composed encoder–predictor sensitivity under a recorded action sequence, not encoder invariance alone. Noise training can improve robustness by reducing same-state encoder neighborhood crossing, reducing rollout amplification along the resulting perturbation directions, or changing how encoder-sensitive directions feed rollout high-gain directions. It need not reduce the rollout-side Jacobian uniformly. Small encoder shifts can still be amplified by $J_{\bar{G}_\mathbf{a}}$, while large nuisance shifts can become harmless after rollout if the product $J_{\bar{G}_\mathbf{a}} J_E$ is small in those directions. The measured state-proxy margin pass event is

$$M = \mathbf{1}[d_{\text{proxy diff}} > d_{\text{same state noise}} + \delta_m] \,,$$

with $\delta_m = 0.10$ in the frozen protocol. It is a finite state-proxy consistency criterion, not a full semantic-label or action-faithfulness guarantee.

The theoretical statements in this section are intended as diagnostic orientation rather than as closed-loop control guarantees. They identify the failure modes that a paired clean/noisy diagnostic should measure: rollout-disagreement tails, clean candidate margins, and complete proxy collapse. They do not assert that a low empirical ATR is a uniform bound for adaptive planning, repeated replanning, environment-feedback stability, or preservation of task semantics.

3.6 Operational metrics

The reported diagnostic uses two metrics matched to this radius–margin logic. ACPC Tail Risk (ATR) is the checkpoint quantile of the canonical per-anchor radius in Equation (9). The default protocol uses $H = 8$, uniform $\alpha_k = 1/H$, the per-anchor clean observed-transition q50 scale in Equation (8), a mean over noise draws within each anchor, and then the checkpoint q90 over anchors; lower is better. The q90 is a fixed empirical reporting choice, not a theoretical constant. Selective Margin Pass Rate (SMPR) is the pass rate of programmatic label-crossing pairs whose clean different-state rollout distance exceeds the q90 same-state noisy tube by normalized margin 0.10; higher is better.

The frozen gate is therefore not ATR alone. ATR tracks the radius side, while SMPR checks whether the selected proxy pairs remain outside the perturbation tube. Low ATR without high SMPR is not accepted because a constant encoder–predictor also makes clean and corrupted views agree. However, the MVE controls show that the selected labels and actions add no established increment over random-label or action-permuted controls. SMPR is consequently retained as a collapse/proxy guard, not evidence of semantic or action sufficiency.

Collapse caveat. Low same-state ACPC alone is insufficient. A constant encoder–predictor can make every clean and perturbed rollout identical while mapping all states to identical projected futures. The implemented positive-margin SMPR rejects this complete collapse in all four MVE rows. It does not respond before the final point of a scale-invariant progressive-collapse path, so stronger anti-collapse sensitivity remains open.

4 Experimental Setup

The protocol separates behavior endpoints from diagnostics. Closed-loop success under Gaussian evaluation noise establishes the behavioral phenomenon; ATR and SMPR are then computed on fixed checkpoints to localize whether recovered endpoints also satisfy the selective ACPC pattern. Training seed, checkpoint epoch, evaluation seeds, and trajectory counts are kept fixed across the main comparisons.

4.1 Models and noise intervention

LeWM [20] is an end-to-end JEPA world model trained with a latent prediction loss and SIGReg regularization for latent-space model-predictive control. PushT (2D pushing), TwoRoom (2D navigation), Reacher (2D arm), and Cube (3D manipulation) form a heterogeneous stress panel spanning different nuisance/discriminability regimes. We add per-frame Gaussian noise to the input sequence as a full-sequence augmentation: each frame is independently noised with probability $p = 1.0$, and the standard deviation is drawn from Uniform$(0, \sigma_{\max})$. This is a full-sequence input-side augmentation. We sweep $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$ and evaluate observation-only Gaussian noise with the goal image unperturbed as the primary robustness endpoint.

PLDM provides a second inspectable latent-predictive architecture with the same four task interfaces. We apply the same paired-history, projection, normalization, and frozen scoring code without retraining a probe or changing stressor-specific thresholds. Only one independently trained PLDM family is available, so this establishes implementation-level analysis portability and an exploratory relative check, not training-seed-replicated model-family generalization. The off-axis stressors are fixed Gaussian blur ($k = 15$) and resize (0.25); both are evaluated for every base/endpoint task–training-seed block rather than selected per outcome.

4.2 Evaluation and diagnostics

The canonical LeWM development grid contains 36 epoch-10 checkpoints for training seed 3072: four tasks and, for each task, a no-noise baseline plus eight noise-trained checkpoints. Each checkpoint is evaluated with three evaluation seeds (42/43/44) and 100 trajectories per seed; canonical tables report the population mean ± population std across those conditional evaluation replicates. Independent LeWM training seeds 3073 and 3074 use the same nine-point sweep. Together they provide three training seeds for the main Gaussian endpoint in Figure 1.

CAL contains only LeWM seed3072. It fixes $H = 8$, uniform horizon weights, five paired noise draws over 100 anchors, ATR q90, the SMPR q90 tube/35% neighborhood/margin-0.10 rule, and thresholds $\tau_{\mathrm{ATR}} = 1.1783$, $\tau_{\mathrm{SMPR}} = 0.9508$. The final protocol file is immutable with SHA-256 `edcb801c3da388e673c9b55d706a558aa01da7a281fc151e52e1cda566045a21`. E1 (LeWM seeds3073/3074), E2 (one PLDM training seed), E3 (fixed-ρ blur/resize), and E4 (failed target-view repair) prohibit threshold search. Behavior is joined after frozen diagnostic scoring in the aggregate validation scripts; this is not claimed as operator blinding of every raw diagnostic run.

E3 retains all 32 base-to-endpoint pairs: 24 LeWM rows (four tasks, three independently trained seeds, two stressors) and eight PLDM rows (four tasks, one training family, two stressors). The positive-transfer label is fixed before diagnostic results are joined: stressed success must improve by at least five percentage points while clean success drops by at most five. We report two estimands without stressor-specific tuning: the absolute Gaussian-frozen gate on each endpoint and the natural paired rule $\Delta S > 0$. Uncertainty for the LeWM paired result resamples the 12 task–training-seed blocks 5,000 times while retaining blur and resize together inside each block. Evaluation seeds 42/43/44 remain conditional measurement replicates, not independent training runs.

The results follow a three-layer evidence chain. Closed-loop evaluation remains the behavioral authority. Absolute screening tests whether a family-calibrated gate transfers to held-out seeds or stressors, while paired comparison tests whether diagnostic improvement tracks robustness improvement relative to a reference checkpoint. Fixed-pool certificates, local sensitivity, component baselines, and guard counterfactuals then determine what mechanism can be attached to those empirical associations.

5 Experiments

The experiments use additive Gaussian pixel noise as the controlled development stressor. We first establish the behavioral failure and matched-stressor recovery, then test held-out LeWM calibration, analysis-interface portability to PLDM, and within-LeWM paired comparison across fixed blur/resize stressors. The study does not claim a cross-family absolute score, a black-box diagnostic, superiority over robust-training methods, or general corruption coverage beyond the two fixed off-axis stressors.

All results use the protocol in Section 4: LeWM-base denotes the no-noise checkpoint, and LeWM+noise denotes the eight-level $\sigma_{\max}$ sweep.

5.1 Gaussian observation noise breaks no-noise checkpoints

Observation-only Gaussian noise sharply degrades all four no-noise checkpoints, with the largest losses on PushT and Reacher. Table 2 reports the success rates: each training-seed value is first averaged over the three evaluation seeds, and the table then reports mean ± population std across the three training seeds. The goal image is kept clean in every column.

Table 2: Observation-only Gaussian noise sharply degrades no-noise LeWM checkpoints across all four tasks. Values are mean ± population std across training seeds 3072/3073/3074; each training-seed value averages evaluation seeds 42/43/44 with 100 trajectories per evaluation seed. The goal image is kept clean.

Task	eval $\sigma = 0$	eval $\sigma = 0.03$	eval $\sigma = 0.05$	eval $\sigma = 0.08$
TwoRoom	94.67 ± 0.72	92.56 ± 1.23	81.33 ± 6.77	68.78 ± 2.45
PushT	81.78 ± 4.89	53.56 ± 12.91	19.22 ± 9.12	7.22 ± 5.81
Reacher	59.22 ± 1.03	42.56 ± 2.99	33.56 ± 1.77	18.22 ± 0.42
Cube	65.22 ± 1.10	63.00 ± 1.19	46.67 ± 1.96	43.11 ± 3.27

LeWM-base is strong on unperturbed images, but observation noise alone produces large closed-loop losses at $\sigma = 0.08$: PushT loses 74.56 ± 5.55 points, Reacher 41.00 ± 1.44 points, TwoRoom 25.89 ± 2.38 points, and Cube 22.11 ± 2.78 points between the first and last columns. The progression across noise levels is smooth enough to show that the endpoint is not an isolated artifact of one severity setting. PushT degrades most sharply, consistent with contact-heavy continuous control being sensitive to visual precision.

5.2 Full-sequence Gaussian augmentation yields task-dependent recovery bands

Full-sequence Gaussian noise training produces broad, task-dependent recovery bands rather than a stable universal checkpoint ranking. Figure 1 shows the full sweep aggregated across training seeds 3072/3073/3074; each plotted point first averages evaluation seeds 42/43/44 within a training seed and then reports mean and population std across the three training seeds.

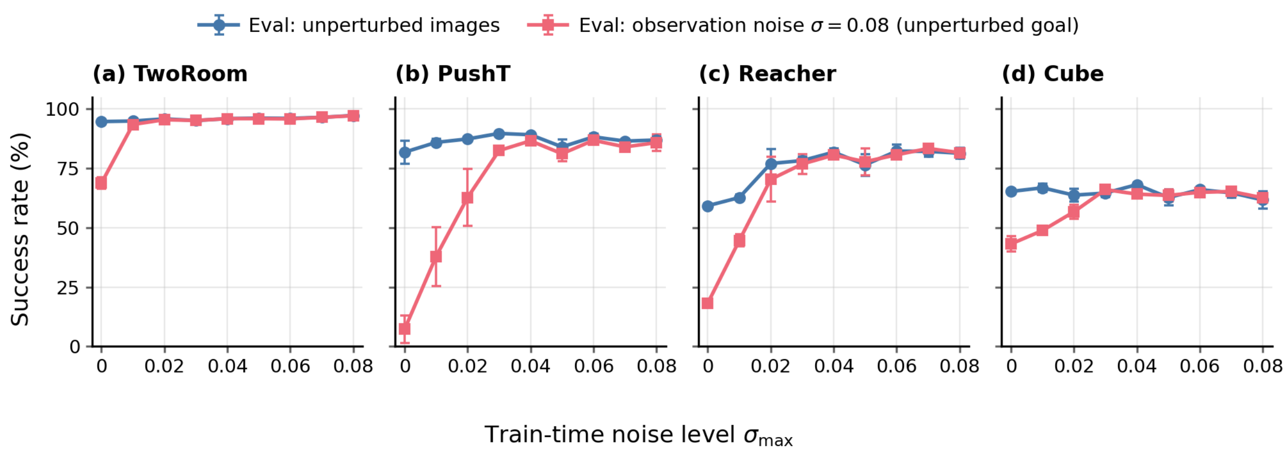

Figure 1: Full-sequence Gaussian noise training produces broad, task-dependent recovery bands rather than a universal checkpoint ranking. Each point averages evaluation seeds 42/43/44 within each training seed and then reports mean across training seeds 3072/3073/3074; error bars show population std across training seeds. Blue circles show unperturbed evaluation; red squares show observation-only Gaussian noise $\sigma = 0.08$ with an unperturbed goal.

Because Figure 1 already aggregates the full sweep across three training seeds, we do not report a separate endpoint leaderboard in the main text. The fixed $\sigma_{\max} = 0.08$ endpoint substantially improves the observation-noise endpoint on TwoRoom, PushT, and Reacher, with a weaker positive Cube signal. Exact all-seed sweep values are reported in Appendix C for reproducibility.

A three-seed blur/resize evaluation later in this section serves as an auxiliary severe-stressor evaluation outside the matched-Gaussian endpoint. The next subsection asks whether the recovered Gaussian endpoints also move in the ATR/SMPR diagnostics.

5.3 Paired radius/guard co-movement and its limits

Recovered checkpoints move in both frozen diagnostic components: the same-state predictive-radius tail contracts and the state-proxy guard pass rate rises. ATR is the q90 across per-anchor canonical weighted-stacked horizon radii after each anchor's noise draws are averaged; every radius is normalized by that anchor's clean observed-transition q50 scale. SMPR uses a q90 same-state tube, a closest-35% state neighborhood, and positive normalized margin 0.10. Both are computed without retraining on the fixed no-noise and $\sigma_{\max} = 0.08$ checkpoints for training seeds 3072/3073/3074. Figure 2 shows the paired endpoint movement, with exact population mean/std values reported in Appendix D.

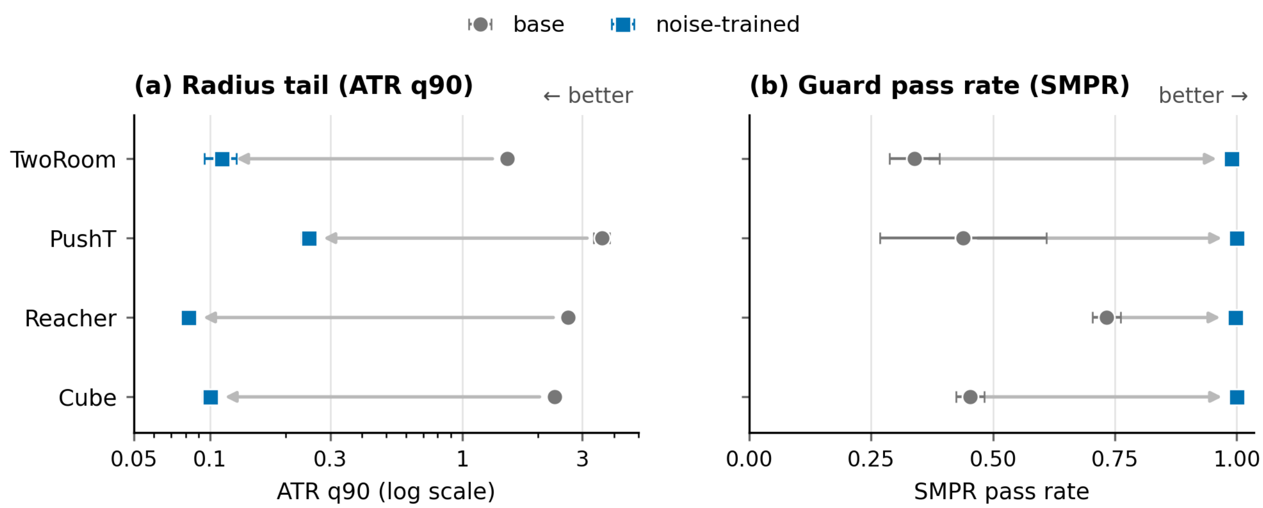

Figure 2: Noise-trained endpoints contract the ATR radius tail and increase the positive-margin state-proxy SMPR on all four tasks. ATR is the q90 across per-anchor weighted-stacked same-state rollout radii, with per-anchor clean-transition q50 normalization and within-anchor mean aggregation over noise draws; SMPR uses normalized margin 0.10. Points report means across training seeds 3072/3073/3074 and horizontal bars report population standard deviations. Co-movement does not establish that the proxy labels or action sequence are necessary; controls are reported in the text and appendix.

The two metrics move in the intended descriptive direction on all four tasks: recovered endpoints have much smaller same-state predictive tails and much higher proxy pass rates. A qualitative PushT neighborhood view is provided in Appendix D.1 as intuition. We interpret post-rollout tightening as a lower composed response to the actual perturbation directions, not as evidence that the rollout-side Jacobian uniformly contracts. Cube remains a boundary case: ATR and SMPR improve strongly, but closed-loop recovery is weaker than PushT or Reacher, so behavior remains the authority.

SMPR uses programmatic state proxies, not oracle semantic annotations: TwoRoom uses doorway/room-side labels, PushT uses T-block pose cells, Reacher uses target/end-effector relation bins, and Cube uses cube-pose/goal-relation cells. Controls on a TwoRoom+PushT base/endpoint MVE correctly reject a constant collapsed rollout in all 4/4 rows and accept an identical clean/noisy positive control in 4/4. They also give exactly zero SMPR change under action permutation, no pre-constant response along a scale-invariant progressive-collapse path, and only a 0.026 mean advantage of task-derived over random labels that is not strict on all rows. Thus the guard establishes complete-collapse rejection and proxy-level consistency, not action relevance or semantic sufficiency. TwoRoom uses an environment-geometry proxy and PushT a sequence-goal state-derived proxy in the one-seed MVE, giving SMPR 0.156 → 1.000 and 0.276 → 1.000, respectively. PushT's HDF5 lacks the simulator goal state, so even this row is not called a simulator oracle.

5.3.1 Full-sweep and held-out validation

Across all $4 \times 3 \times 9 = 108$ task–training-seed–checkpoint rows, recovered portions of the Gaussian sweep consistently occupy lower normalized ATR and higher SMPR than fragile rows. Figure 3 places the closed-loop observation-noise score above the two diagnostic traces so behavioral recovery and diagnostic movement remain visually distinct. Fixed-pool top-1 disagreement and proxy-gap traces are kept in Appendix E, where they support the planner-side radius–margin audit.

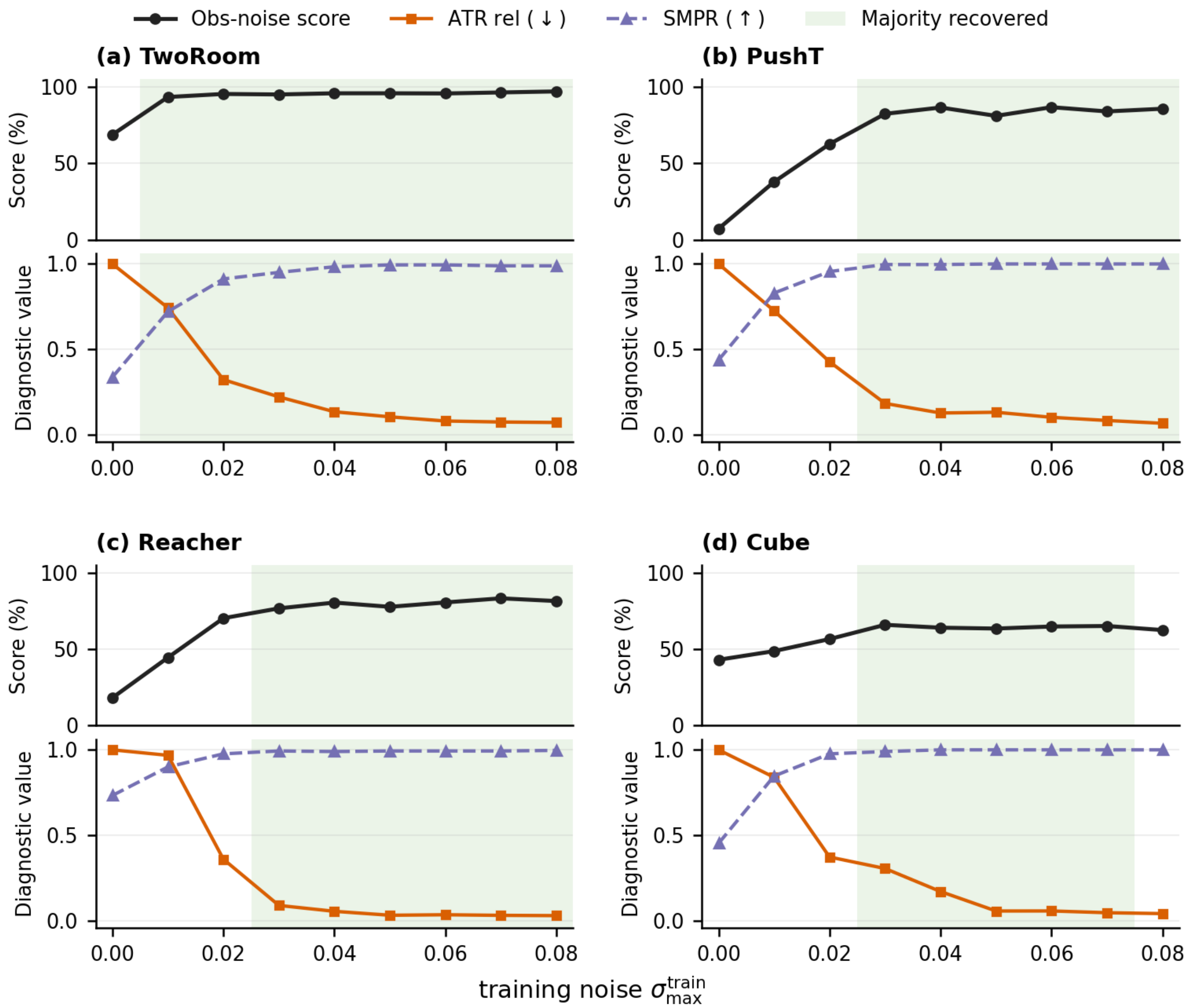

Figure 3: Recovered rows align with lower ATR and higher positive-margin proxy SMPR across the Gaussian training grid. Each task block averages training seeds 3072/3073/3074 at the same $\sigma_{\max}$ value and separates the observation-noise score (upper axis) from normalized ATR and SMPR at margin 0.10 (lower axes; ATR ↓, SMPR ↑). Continuous green bands mark training-noise ranges where a majority of seed rows meet the behavioral recovery criterion.

5.3.2 Frozen external validation and component baselines

The public-v1 thresholds were selected only on the 36-row LeWM seed3072 Gaussian calibration split (CAL), then applied read-only with protocol SHA-256 `edcb801c3da388e673c9b55d706a558aa01da7a281fc151e52e1cda5 66045a21`. Table 3 reports the *absolute-screening* tests. E1 strongly confirms calibration portability to two held-out LeWM training seeds. E2 executes the same scoring pipeline on one PLDM training family, but 12 false positives and an all-positive Reacher branch show that LeWM numerical thresholds are not a shared cross-family calibration. E3 fixes training strength at $\rho = 0.08$ and pre-fixes the strongest blur/resize mapping; its low recall

Table 3: Read-only absolute-screening validation using the LeWM Gaussian-frozen gate. E2 tests shared numerical calibration on one PLDM training family, not whether the analysis interface can be executed. E3 contains 16 pre-mapped fixed endpoints; the 32-pair relative analysis is reported separately.

Split	rows	balanced acc.	AUPRC	boundary
E1 held-out LeWM seeds	72	0.945	0.965	two independent training seeds
E2 PLDM	36	0.684	0.541	one family; calibration not shared
E3 fixed-ρ blur/resize	16	0.563	0.737	conservative: recall 0.375
E4 failed repair	64	0.907	0.587	target branch has no positives

shows that the absolute gate is conservative off-axis. The all-pairs relative E3 estimand is reported separately in Section 5.6. E4 is a falsification split: all 32 target-view rows are behavior-negative and rejected, while the paired full-sequence branch retains false-pass rate 0.727.

Component baselines prevent attributing the result to the eight-step, correctly action-conditioned map. On the combined E1+E2 rows, the joint gate has the highest balanced accuracy (0.822), but encoder-only and H1 both reach 0.793, action-zeroed H8 reaches 0.805, and correct-action H8 reaches only 0.756. Their AUPRC values are 0.792, 0.816, 0.841, and 0.828, respectively. Action-shuffled H8 also reaches AUPRC 0.838. Hence neither long-horizon rollout nor correct-action conditioning is empirically necessary in this benchmark. SMPR and the conjunction remain useful frozen features, but their increment is metric-dependent rather than a demonstrated causal mechanism.

Table 4: CAL-frozen diagnostic baselines on the combined E1+E2 external rows. Thresholds are never selected on external behavior; clean score is not training-free.

Diagnostic	Balanced accuracy	AUPRC
encoder_q90	0.793	0.792
h1_q90	0.793	0.816
action_shuffled_h8_q90	0.780	0.838
action_zeroed_h8_q90	0.805	0.841
time_shuffled_h8_q90	0.756	0.831
atr_h8_q90	0.756	0.828
clean_score	0.630	0.816
smpr	0.785	0.809
atr_smpr_joint	0.822	0.811

The Gaussian training strength ρ is evaluated only in a privileged confound appendix and is excluded from external diagnostic leaderboards. ATM is also not approximated by these baselines: official ATM trains fresh inverse probes across real and predicted transition domains and fits within-task screening coefficients [4]; reproducing it requires a new probe-training split outside this frozen release.

5.4 Planner-side radius–margin audit

The candidate-wise audit recomputes all 108 LeWM checkpoints, 100 histories per checkpoint, and the same ordered pool of $K = 65$ candidates on clean and corrupted branches. The legacy coarse condition uses the maximum drift and clean top-1/top-2 margin. The sharp condition in Equation (13) instead compares every competitor's clean gap with its own drift and the clean winner's drift; it directly matches the deterministic top-1 implication.

Across 10,800 sampled histories, coarse coverage is 0.478, sharp coverage is 0.700, and the observed fixed-pool top-1 flip rate is 0.208. Among sharp-certificate failures the flip rate is 0.694. A hierarchical bootstrap over the 12 task–training-seed blocks gives 95% intervals $[0.438, 0.515]$, $[0.679, 0.723]$, $[0.193, 0.227]$, and $[0.664, 0.727]$ for these four quantities. The sharp condition has zero certified flips by construction of the deterministic implication; this invariant check confirms implementation consistency but is not treated as independent statistical evidence.

The curve is descriptive selection by sharp slack, not learned probability calibration. A K-sensitivity audit at $K \in \{8, 16, 32, 65\}$ is retained in the artifact, and block-level coverage is reported in Appendix E.2. Earlier q50/q90 recovery-band overlays remain historical mechanism views; the sharp candidate-wise condition is the paper's exact fixed-pool result.

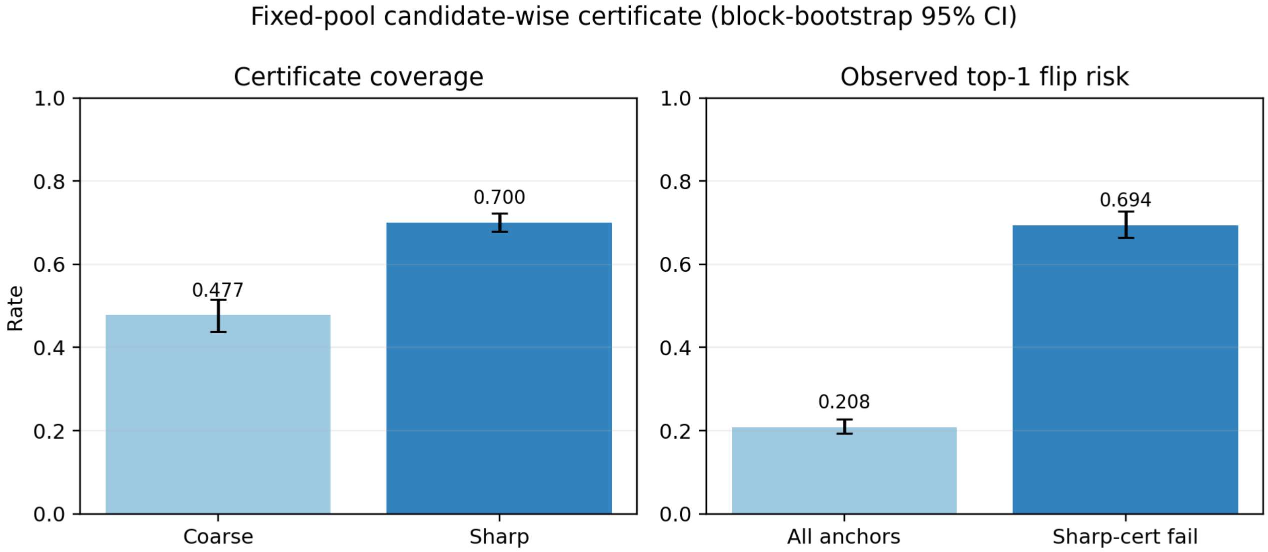

Figure 4: Candidate-wise fixed-pool calibration over 108 checkpoints and 10,800 histories. The sharp slack produces a risk–coverage curve: observed flip rate is 0.208 at full coverage, 0.022 at 75% coverage, and zero on the 70.0% satisfying the exact sharp condition. Coverage/risk is for the sampled ordered pool and does not transfer automatically to adaptive CEM or repeated replanning.

Table 5: Matched-map Gaussian linearization calibration. Ratios compare measured $\mathbb{E}[R^2]/\sigma^2$ with the covariance-aware exact-JVP/Hutchinson trace at $\sigma = 0.0025$; relative error averages all four small-σ probes. Values are medians or means over three training seeds. The remainder order is descriptive and no global guarantee is implied.

Task	base ratio	endpoint ratio	mean rel. error (base/end)
TwoRoom	0.94	1.93	0.35/0.46
PushT	0.85	1.22	0.30/0.13
Reacher	1.35	1.18	5.23/0.16
Cube	1.28	1.16	2.89/0.07

5.5 Local calibration supports contraction but exposes nonlinearity

The matched-map audit covers every task/training-seed base, recovery-onset, and endpoint checkpoint (36 rows). It uses the same 16 fixed anchors and unnormalized $H = 8$ weighted-stacked map for finite differences and exact-autograd JVPs. Eight Rademacher probes are scaled by reciprocal ImageNet channel standard deviation so their covariance matches raw pixel-space Gaussian noise; eight Gaussian draws are evaluated at $\sigma \in \{0.0025, 0.005, 0.01, 0.02\}$. This corrects both the horizon-scale and input-coordinate mismatch that would arise from comparing an unweighted rollout norm or normalized-input isotropic trace with the pixel-space stressor.

The covariance-aware JVP trace falls at the endpoint to 0.0027, 0.0087, 0.0065, and 0.0287 of the within-task base median for TwoRoom, PushT, Reacher, and Cube. At the smallest σ, measured/JVP ratios are reasonably close to one for most aggregates, but the TwoRoom endpoint median is 1.93 and mean relative error across the four probes reaches 5.23 and 2.89 for base Reacher and Cube. Descriptive log–log remainder slopes are not systematically three. Thus the audit supports a strong local composed-sensitivity reduction while rejecting a claim that the tested finite range uniformly validates the $O(\sigma^3)$ approximation.

Horizon/quantile sensitivity reaches the same qualitative conclusion as the component baselines. At q90, endpoint/base ATR ratios for $H = 1$ versus $H = 8$ are 0.05/0.06 (TwoRoom), 0.06/0.09 (PushT), 0.02/0.02 (Reacher), and 0.04/0.04 (Cube). Varying q80/q90/q95 likewise changes magnitudes little. We therefore keep $H = 8$ as the preregistered public-v1 definition but remove any claim that long-horizon rollout is empirically necessary here.

Appendix F retains the raw encoder, rollout, and composed JVP decomposition and both κ definitions. The rollout-side trace is task-dependent rather than uniformly smaller, so no standalone predictor-Jacobian repair is claimed.

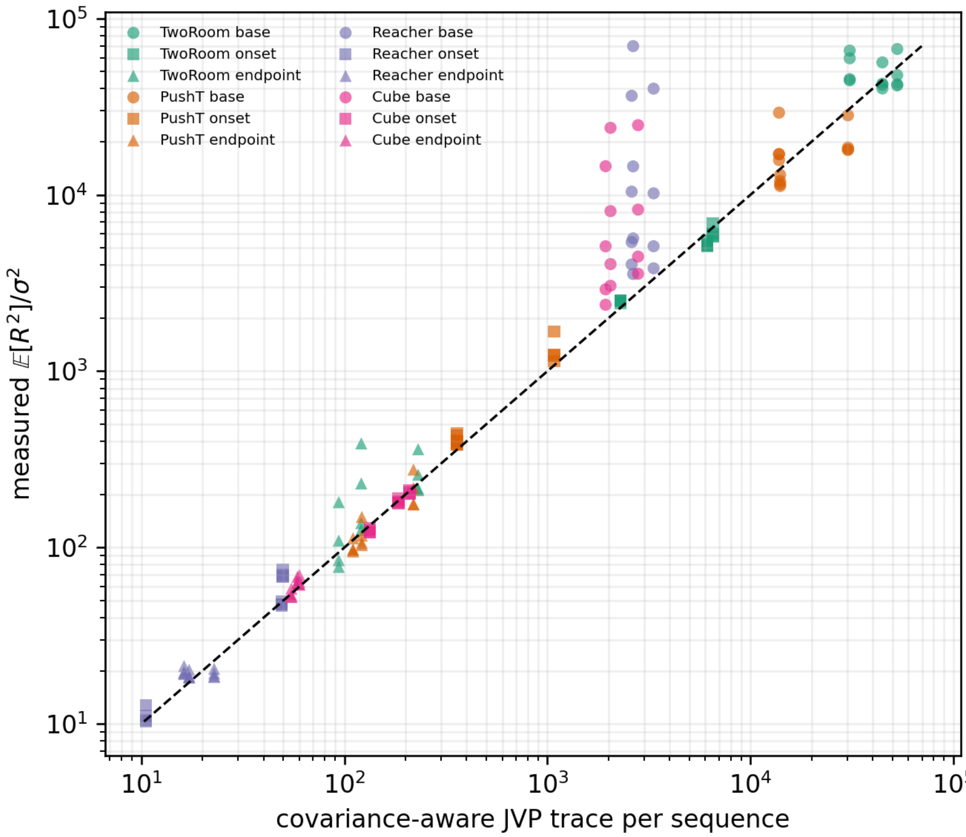

Figure 5: Measured $\mathbb{E}[R^2]/\sigma^2$ versus the covariance-aware JVP/Hutchinson trace per sequence for all 36 checkpoints and four small noise levels. The dashed line is equality. Dispersion, especially for base Reacher/Cube at larger small-σ, is retained as a calibration limit rather than hidden behind endpoint/base direction.

Table 6: Horizon/quantile sensitivity at fixed pixel-noise $\sigma = 0.08$. Each entry is the median endpoint/base horizon-v2 ATR ratio over three training seeds (lower is a larger endpoint reduction). Horizon and quantile effects are descriptive; they do not retune the frozen gate.

	$q = 0.90$ by horizon				$H = 8$ by quantile		
Task	H1	H2	H4	H8	q80	q90	q95
TwoRoom	0.05	0.04	0.05	0.06	0.06	0.06	0.06
PushT	0.06	0.07	0.06	0.09	0.07	0.09	0.09
Reacher	0.02	0.03	0.03	0.02	0.02	0.02	0.02
Cube	0.04	0.03	0.04	0.04	0.03	0.04	0.04

5.6 Within-family paired changes transfer across blur and resize

The fixed-strength E3 audit evaluates both blur and resize for every base/endpoint block: 24 LeWM pairs and eight exploratory PLDM pairs. Diagnostic artifacts are scored before behavior is joined, severity is fixed, and neither the Gaussian-frozen absolute gate nor the paired zero threshold is retuned. Table 7 separates the two uses.

Table 7: Absolute screening and paired comparison are distinct uses. The absolute row applies the LeWM Gaussian-frozen joint gate to all LeWM blur/resize endpoints. Paired rows use one untuned rule, Δjoint score > 0, for both stressors. PLDM contains one training family and is exploratory. ρ_s correlates continuous diagnostic and stressed-success changes.

Use	scope	n	BA	AUPRC	precision	recall	ρ_s
absolute	LeWM blur+resize	24	0.733	0.856	1.000	0.467	0.530
paired $\Delta > 0$	LeWM blur+resize	24	0.889	0.974	0.882	1.000	0.864
paired $\Delta > 0$	LeWM blur	12	0.875	0.953	0.889	1.000	0.860
paired $\Delta > 0$	LeWM resize	12	0.900	1.000	0.875	1.000	0.762
paired $\Delta > 0$	PLDM blur+resize	8	0.833	1.000	0.500	1.000	0.667

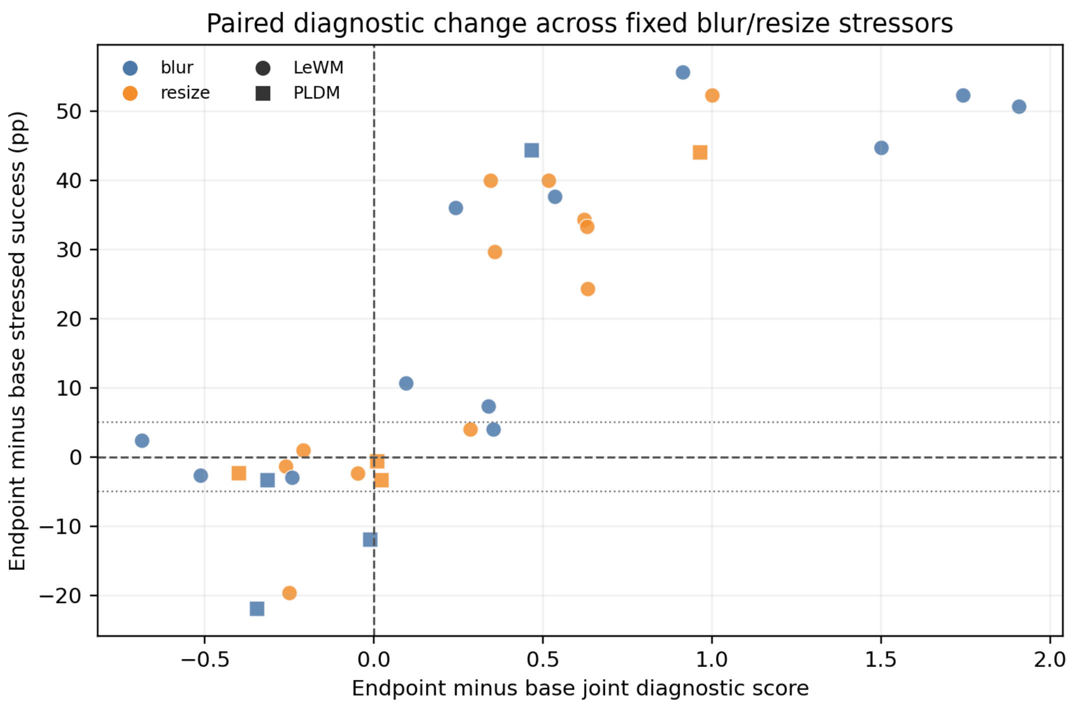

Figure 6: Paired diagnostic improvement tracks closed-loop robustness improvement across fixed blur and resize stressors. Each point is one base-to-endpoint task–training-family pair; circles are three LeWM training seeds and squares are one PLDM training family. Colors identify stressors, not outcome classes. The dashed axes show the single $\Delta S = 0$ decision boundary and zero behavioral change; dotted horizontal lines show the pre-specified ± 5-point behavioral band.

On all 24 LeWM pairs, the shared rule $\Delta S > 0$ reaches balanced accuracy 0.889, AUPRC 0.974, precision 0.882, and recall 1.000. It remains consistent when split without retuning: balanced accuracy is 0.875 for blur and 0.900 for resize. Continuous diagnostic and stressed-success changes have Spearman correlation 0.864 and signed agreement 0.917. A 5,000-repeat bootstrap over the 12 task–training-seed blocks, retaining both stressors within each block, gives 95% intervals $[0.700, 1.000]$ for balanced accuracy, $[0.869, 1.000]$ for AUPRC, $[0.641, 0.932]$ for Spearman correlation, and $[0.792, 1.000]$ for signed agreement.

Across all 32 rows, the paired association is 0.871. The one-family PLDM slice gives balanced accuracy 0.833 and Spearman correlation 0.667, but it contains only eight rows and two positives. We therefore use PLDM to demonstrate that the analysis interface can be instantiated on another latent predictive model, not to claim training-seed-replicated cross-model performance or a shared absolute threshold.

The paired signal is also not unique to correct-action H8. Across the 32 rows, Spearman correlations are 0.865 for encoder change, 0.862 for correct-action H8, 0.874 for time-shuffled H8, and 0.871 for the joint score.

This supports a family of paired latent-sensitivity diagnostics; the rollout/candidate construction supplies the control-facing interpretation, not an empirical complexity advantage.

Absolute screening answers a different question. On the 24 LeWM pairs, the unchanged Gaussian-frozen gate has balanced accuracy/AUPRC 0.733/0.856, precision 1.000, and recall 0.467: a pass is reliable in this sample, but many behavior-positive endpoints are missed. Thus the supported cross-stressor result is reference-based relative comparison within a calibrated family, while single-checkpoint absolute screening remains conservative.

6 Discussion and limitations

What the evidence supports. The central result is a hierarchy rather than a universal score. The paired-radius/guard analysis interface is implemented for LeWM and PLDM without changing its mathematical objects. Within LeWM, an absolute gate calibrated on one Gaussian seed transfers strongly to two held-out training seeds and conservatively to blur/resize, while a single reference-based zero threshold tracks relative robustness changes across both off-axis stressors. The sharp candidate-wise condition gives exact sampled-pool implications and measured coverage; covariance-aware JVP/finite-difference calibration supports local composed-sensitivity contraction. Block uncertainty, failed repairs, and component controls are retained as part of the evidence rather than selected away.

Scope. Analysis-interface portability is conditional on access to paired views, the model representation/predictor, and compatible projections; it is not a black-box theorem for arbitrary agents. Absolute thresholds remain family-calibrated. The paired zero-threshold result requires a reference checkpoint and stressor-specific diagnostic recomputation, and it covers only blur and resize at one fixed severity. PLDM has one training family; evaluation seeds 42/43/44 are conditional measurement replicates. The sharp condition covers a once-sampled ordered pool; it is not an adaptive-planning guarantee for CEM resampling, repeated replanning, or environment feedback. Encoder/H1/action/time controls show no empirical necessity for a long horizon or correct-action conditioning. SMPR rejects complete collapse but does not establish task-label, action-relevance, progressive-collapse, or four-task oracle sufficiency. We do not claim superiority over robust-control training methods.

What remains. Broader empirical portability requires independently trained PLDM seeds or additional model families, more corruption families and severities, and stronger semantic/planner guards. An official ATM comparison requires fresh real/predicted transition probes and a preregistered probe split; current baselines are not relabeled as ATM. Adaptive-CEM guarantees require a separate analysis of resampling and feedback. These are extensions of the bounded claim, not missing premises silently assumed by the present result.

7 Conclusion

This paper separates the portability of a paired latent robustness analysis from the portability of its numerical calibration. The same radius/guard interface is instantiated on LeWM and PLDM, but LeWM thresholds are not promoted to cross-family constants. Within LeWM, the absolute gate transfers strongly across training seeds and acts as a high-precision, low-recall blur/resize screen. More importantly for comparing checkpoints, one untuned $\Delta S > 0$ rule is shared across blur and resize: it reaches balanced accuracy/AUPRC 0.889/0.974 and tracks closed-loop robustness change with Spearman correlation 0.864 across 24 pairs.

The strongest exact result remains candidate-wise: positive sharp slack guarantees top-1 stability for the evaluated fixed pool and covers 70.0% of 10,800 histories. Local JVP calibration supports composed-sensitivity contraction, while encoder/H1/action/time controls show that the empirical diagnostic signal is not uniquely long-horizon or action-specific. The resulting contribution is a portable analysis framework with family-calibrated absolute use, a strongly supported within-family paired cross-stressor use, and explicit fixed-pool and semantic boundaries.

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A Proofs

These proofs support the diagnostic use of ATR and SMPR; they do not provide closed-loop control guarantees.

Proposition 2 (ACPC controls candidate-cost drift)**.** *Assume that J is locally L_J-Lipschitz on the clean/perturbed projected-rollout neighborhood evaluated by the fixed candidate set under metric d_H. If*

$$d_H\big(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a})), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}))\big) \leq \epsilon,$$

then

$$|C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \leq L_J \epsilon.$$

Proof. The planner cost J is L_J-Lipschitz on the projected-rollout sequence, so the cost difference is bounded by L_J times the clean/perturbed projected-rollout distance. □

Proposition 3 (Candidate top-1 stability)**.** *Let $\mathcal{A} = \{\mathbf{a}^1, \ldots, \mathbf{a}^K\}$ be the shared candidate set. Assume lower cost is preferred, and let $\mathbf{a}^{(1)}$ and $\mathbf{a}^{(2)}$ denote the lowest- and second-lowest-cost candidates on the clean branch. Suppose that for every candidate j,*

$$|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq \eta.$$

If the clean branch has a strict top-1/top-2 margin

$$\Delta = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g) > 2\eta,$$

then the perturbed branch selects the same top-1 candidate.

Proof. For any candidate $\mathbf{a}^j \neq \mathbf{a}^{(1)}$,

$$C_{\tilde{h}}(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^{(1)}, g) \geq C_h(\mathbf{a}^j, g) - C_h(\mathbf{a}^{(1)}, g) - 2\eta \geq \Delta - 2\eta > 0.$$

Thus every non-top candidate remains more costly than the clean top-1 candidate on the perturbed branch. □

Candidate-wise sharp condition. For clean winner j^* and any competitor j, the drift definition gives

$$C_{\tilde{h}}(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^{j^*}, g) \geq \Delta_j - d_j - d_{j^*}.$$

Therefore positivity of the minimum in Equation (13) makes every perturbed competitor strictly more costly. This proves the deterministic fixed-pool implication without a common worst-case drift bound.

Proof of Theorem 1.

Proof. Let $\mathcal{B} = \{\max_{1 \leq j \leq K} D_j \leq \epsilon\}$. Since each candidate is drawn from q and a single draw has tail probability at most α, the union bound gives

$$\Pr_{\mathcal{A}}(\mathcal{B}^c) \leq \sum_{j=1}^{K} \Pr[D_j > \epsilon] \leq K\alpha.$$

On $\mathcal{B}$, local Lipschitzness of the planner cost gives $|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq L_J \epsilon$ for every candidate. By the top-1 stability proposition, the perturbed branch can choose a different top candidate on $\mathcal{B}$ only when $\Delta_{\mathcal{A}} \leq 2L_J\epsilon$. Therefore the top-1 flip event is contained in

$$\mathcal{B}^c \cup \{\Delta_{\mathcal{A}} \leq 2L_J\epsilon\},$$

which yields the fixed-pool instability bound without any independence assumption beyond the single-candidate tail statement. For the selective diagnostic statement, let $\mathcal{D} = \{M_{\text{diff}} \leq R_\sigma + \delta_m\}$. Selective failure is contained in the union of the fixed-pool planning-instability event and $\mathcal{D}$. Applying the union bound again and using $\Pr(\mathcal{D}) \leq \gamma$ gives the final bound. □

Local Gaussian sensitivity. For small Gaussian pixel noise ξ, a first-order expansion gives $E_\theta(o + \xi) = E_\theta(o) + J_E(o)\xi + \mathcal{R}(\xi)$ with $\|\mathcal{R}(\xi)\| = O(\|\xi\|^2)$. Composing the encoder with the action-conditioned rollout map shows that same-state rollout disagreement measures local sensitivity of the encoder–predictor composition, not encoder invariance alone. ATR reports a high-quantile version of that paired rollout disagreement.

B Experimental Protocol and Artifacts

These details define the fixed evaluation and diagnostic protocol used by the main tables.

LeWM is evaluated on PushT, TwoRoom, Reacher, and Cube using the canonical three evaluation seeds 42/43/44 and 100 trajectories per seed. The main Gaussian robustness endpoint perturbs observation pixels with Gaussian noise at standard deviation 0.08 while keeping the goal image clean. Noise-trained checkpoints use full-sequence input-side Gaussian augmentation with $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$.

ATR is computed at each task–training-seed–checkpoint row from one canonical weighted-stacked horizon radius per anchor. The default is $H = 8$ with uniform $\alpha_k = 1/H$. Each radius is divided by that anchor's q50 clean observed-transition scale, including the transition from the last history observation to the first future observation; multiple noise draws are first averaged within the anchor, and q90 is then taken across anchors. SMPR uses programmatic label-crossing state pairs, q90 tube radius, closest-35% neighborhood, and normalized margin 0.10. The paper reports population mean/std across training seeds 3072/3073/3074.

The legacy field `stepwise_rollout_q90` is retained only for compatibility in appendix artifacts. It pools the $B \times H$ stepwise rollout distances before taking q90; it is not ATR and is not used as the canonical horizon-radius statistic.

C Additional Gaussian Evaluation Tables

These tables report the full observation-only Gaussian evaluation columns available for the three-training-seed sweep behind Figure 1. Each training-seed value first averages evaluation seeds 42/43/44; table cells report mean ± population std across training seeds 3072/3073/3074.

Table 8: Full three-training-seed LeWM observation-only Gaussian sweep evaluation table. Columns report clean evaluation and observation-only Gaussian evaluation at $\sigma = 0.03/0.05/0.08$ with a clean goal image. Values are mean ± population std across training seeds 3072/3073/3074; each training-seed value first averages evaluation seeds 42/43/44 with 100 trajectories per evaluation seed.

Task	training $\sigma_{\max}$	clean eval	obs-only $\sigma = 0.03$ eval	obs-only $\sigma = 0.05$ eval	obs-only $\sigma = 0.08$ eval
TwoRoom	0.00	94.67 ± 0.72	92.56 ± 1.23	81.33 ± 6.77	68.78 ± 2.45
TwoRoom	0.01	94.89 ± 1.10	95.00 ± 0.82	94.78 ± 1.77	93.44 ± 2.08
TwoRoom	0.02	95.78 ± 1.10	95.89 ± 1.29	95.22 ± 1.29	95.44 ± 1.57
TwoRoom	0.03	95.11 ± 0.87	95.00 ± 0.82	95.22 ± 0.83	95.11 ± 1.34
TwoRoom	0.04	95.89 ± 0.42	96.44 ± 0.96	95.78 ± 0.31	95.89 ± 0.57
TwoRoom	0.05	96.11 ± 1.50	96.00 ± 1.91	96.00 ± 1.52	95.89 ± 1.55
TwoRoom	0.06	96.00 ± 1.70	95.56 ± 1.81	95.56 ± 1.40	95.78 ± 1.55
TwoRoom	0.07	96.44 ± 0.63	96.33 ± 0.54	96.11 ± 0.68	96.44 ± 0.31
TwoRoom	0.08	97.22 ± 0.83	97.22 ± 0.68	97.33 ± 0.72	97.11 ± 0.57
PushT	0.00	81.78 ± 4.89	53.56 ± 12.91	19.22 ± 9.12	7.22 ± 5.81
PushT	0.01	85.89 ± 1.50	82.22 ± 3.03	70.33 ± 6.33	37.89 ± 12.37
PushT	0.02	87.33 ± 0.72	85.11 ± 1.13	82.89 ± 2.69	62.67 ± 11.98
PushT	0.03	89.67 ± 0.00	89.67 ± 0.27	88.78 ± 2.08	82.44 ± 1.66
PushT	0.04	89.11 ± 0.83	88.22 ± 1.13	87.33 ± 1.19	86.56 ± 0.16
PushT	0.05	84.00 ± 3.03	84.22 ± 2.53	82.56 ± 3.13	81.11 ± 3.03
PushT	0.06	88.22 ± 1.34	88.89 ± 1.03	87.33 ± 0.47	86.78 ± 0.63
PushT	0.07	86.44 ± 0.87	86.00 ± 1.19	84.22 ± 0.42	84.00 ± 0.27
PushT	0.08	86.89 ± 1.59	86.56 ± 2.20	84.67 ± 2.88	85.78 ± 3.45
Reacher	0.00	59.22 ± 1.03	42.56 ± 2.99	33.56 ± 1.77	18.22 ± 0.42
Reacher	0.01	62.67 ± 1.19	60.67 ± 1.09	56.00 ± 0.72	44.56 ± 2.57
Reacher	0.02	77.00 ± 6.18	76.44 ± 8.17	77.22 ± 6.68	70.33 ± 9.46
Reacher	0.03	78.22 ± 1.66	79.33 ± 0.72	79.11 ± 1.10	76.78 ± 4.17
Reacher	0.04	81.78 ± 1.59	82.00 ± 1.52	81.11 ± 2.62	80.56 ± 1.03
Reacher	0.05	76.44 ± 4.57	77.33 ± 5.76	77.56 ± 4.01	77.78 ± 5.67
Reacher	0.06	82.22 ± 2.79	79.33 ± 2.42	81.67 ± 1.63	80.67 ± 1.19
Reacher	0.07	82.11 ± 2.20	82.44 ± 1.29	81.44 ± 1.85	83.33 ± 1.09
Reacher	0.08	81.33 ± 2.33	82.56 ± 2.08	82.00 ± 1.78	81.56 ± 0.87
Cube	0.00	65.22 ± 1.10	63.00 ± 1.19	46.67 ± 1.96	43.11 ± 3.27
Cube	0.01	66.78 ± 1.81	67.00 ± 2.13	67.56 ± 2.27	48.78 ± 0.31
Cube	0.02	63.67 ± 2.62	61.44 ± 3.45	63.33 ± 1.96	56.67 ± 3.07
Cube	0.03	64.56 ± 1.66	63.44 ± 3.03	65.33 ± 0.94	66.00 ± 1.70
Cube	0.04	68.11 ± 1.03	65.11 ± 1.50	65.44 ± 0.57	64.11 ± 0.83
Cube	0.05	62.67 ± 3.40	63.33 ± 2.42	60.67 ± 3.09	63.56 ± 2.53
Cube	0.06	66.00 ± 1.19	65.89 ± 1.03	64.67 ± 1.09	64.89 ± 1.85
Cube	0.07	64.67 ± 2.16	64.67 ± 1.89	65.44 ± 2.57	65.22 ± 1.55
Cube	0.08	61.67 ± 3.57	61.78 ± 4.16	61.89 ± 2.45	62.56 ± 1.55

D Diagnostic Validation Details

These details report endpoint values, the older matched-Gaussian direction checks, and the new SMPR sensitivity/controls. The endpoint and full-sweep co-movement is descriptive; the component and counterfactual tests in the main text determine which mechanism claims survive.

Table 9: Paired endpoint diagnostics over training seeds 3072/3073/3074. ATR is the q90 normalized same-state clean/noisy rollout disagreement; lower is better. SMPR is the positive-margin ($\delta = 0.10$) programmatic state-proxy pass rate; higher is better. Component and counterfactual controls show that this co-movement does not establish long-horizon, action, or semantic necessity.

Task	ATR base	ATR $\sigma_{\max} = 0.08$	SMPR base	SMPR $\sigma_{\max} = 0.08$
TwoRoom	1.509 ± 0.010	0.111 ± 0.016	0.34 ± 0.05	0.99 ± 0.01
PushT	3.580 ± 0.260	0.247 ± 0.017	0.44 ± 0.17	1.00 ± 0.00
Reacher	2.628 ± 0.048	0.082 ± 0.005	0.73 ± 0.03	1.00 ± 0.00
Cube	2.320 ± 0.045	0.100 ± 0.007	0.45 ± 0.03	1.00 ± 0.00

Table 10: Claim-to-evidence map for the paired robustness analysis. Mathematical interface portability, family calibration, and within-family paired comparison are distinct claims.

Theoretical quantity	Empirical audit	Main observation	Scope
R_σ radius tail	ATR q90 and normalized ATR	Recovered rows occupy low-ATR regions; endpoint ATR falls on all tasks.	q90 is a fixed diagnostic tail summary, not the theorem's calibrated α.
Covariance-aware $J_{G_{\mathbf{a}}} J_E$ trace	Matched-map finite differences and exact JVP/Hutchinson	Endpoint/base trace ratios are 0.0027/0.0087/0.0065/0.0287 across the four tasks.	Finite-range calibration is nonlinear for base Reacher/Cube; no global or closed-loop guarantee.
Candidate-wise sharp slack	10,800-history fixed-pool audit	Sharp coverage is 0.700 versus coarse coverage 0.478; observed overall flip rate is 0.208.	Exact only for the evaluated ordered pool; certified-zero flips are an invariant, not independent evidence.
Candidate instability	Sharp-slack risk–coverage curve	Observed flip rate falls to 0.022 at 75% coverage and zero at exact sharp-pass coverage.	Descriptive selection by slack, not probability calibration or adaptive CEM.
Complete proxy collapse	Positive-margin SMPR controls	Constant collapse is rejected in 4/4 MVE rows; the identical clean/noisy positive control passes 4/4.	Action permutation has zero effect and task-label increment is not established.
Analysis-interface portability	Identical paired-radius/guard pipeline on LeWM and PLDM	Both architectures produce complete paired diagnostic artifacts without retraining a probe.	Requires inspectable representations/predictions; PLDM has one training family.
Family-calibrated absolute screening	E1/E2 and unchanged LeWM blur/resize gate	E1 reaches 0.945/0.965; LeWM blur/resize precision/recall are 1.000/0.467; PLDM shared calibration is weak.	Thresholds are not claimed to be cross-family constants.
Within-family paired comparison	All 24 LeWM blur/resize base-to-endpoint pairs	One $\Delta S > 0$ rule reaches 0.889/0.974 and Spearman 0.864.	Requires a reference checkpoint; two stressors at one severity, not universal corruption transfer.

Table 11: ATR, proxy SMPR, and fixed-pool top-1 flip co-movement across the Gaussian sweep. Rows are split by the closed-loop recovery-band label. SMPR is a positive-margin state-proxy guard; controls do not establish incremental task-label or action relevance.

Task	ATR fragile $\to$ recovered	SMPR fragile $\to$ recovered	top-1 flip fragile $\to$ recovered
TwoRoom	$1.00 \to 0.12$	$0.36 \to 0.98$	$0.79 \to 0.07$
PushT	$0.72 \to 0.10$	$0.85 \to 1.00$	$0.38 \to 0.06$
Reacher	$0.97 \to 0.04$	$0.90 \to 0.99$	$0.54 \to 0.02$
Cube	$0.71 \to 0.07$	$0.93 \to 1.00$	$0.53 \to 0.01$
ALL	$0.84 \to 0.09$	$0.86 \to 1.00$	$0.54 \to 0.04$

Table 12: All nine behavioral-label settings preserve the expected ATR/SMPR directions in all four tasks. Each cell reports ATR-pass, SMPR-pass task counts for the indicated clean-score tolerance.

	clean-score tolerance		
Recovery fraction	3	5	10
0.70	4/4, 4/4	4/4, 4/4	4/4, 4/4
0.80	4/4, 4/4	4/4, 4/4	4/4, 4/4
0.90	4/4, 4/4	4/4, 4/4	4/4, 4/4

Table 13: SMPR V1 sensitivity summaries. Distribution parameters use all 36 CAL checkpoints; pair-selection parameters use the two-task four-checkpoint MVE.

Factor	Level	Mean SMPR	Mean pair count
noise_draws	1	0.866	89.8
noise_draws	5	0.872	89.8
radius_quantile	0.8	0.899	89.8
radius_quantile	0.9	0.872	89.8
radius_quantile	0.95	0.838	89.8
margin_delta_norm	0.0	0.882	89.8
margin_delta_norm	0.05	0.876	89.8
margin_delta_norm	0.1	0.872	89.8
margin_delta_norm	0.25	0.852	89.8
local_state_quantile	0.1	0.570	14.0
local_state_quantile	0.25	0.616	21.0
local_state_quantile	0.35	0.639	24.0
local_state_quantile	0.5	0.640	26.0
label_binning	median	0.639	24.0
label_binning	quartile	0.618	29.5
label_binning	fixed_physical	0.621	31.0

Table 14: SMPR controls on the TwoRoom+PushT base/endpoint MVE.

Control	Rows	Mean SMPR	Joint passes
action_permutation	4	0.639	2
constant_collapsed_rollout	4	0.000	0
global_far_neighbor	4	0.625	2
identical_clean_noisy_positive	4	1.000	4
progressive_collapse	16	0.639	8
same_label_nearest_neighbor	4	0.587	2
state_label_permutation	4	0.613	2
task_grounded_default	4	0.639	2

D.1 Qualitative ACPC neighborhood view

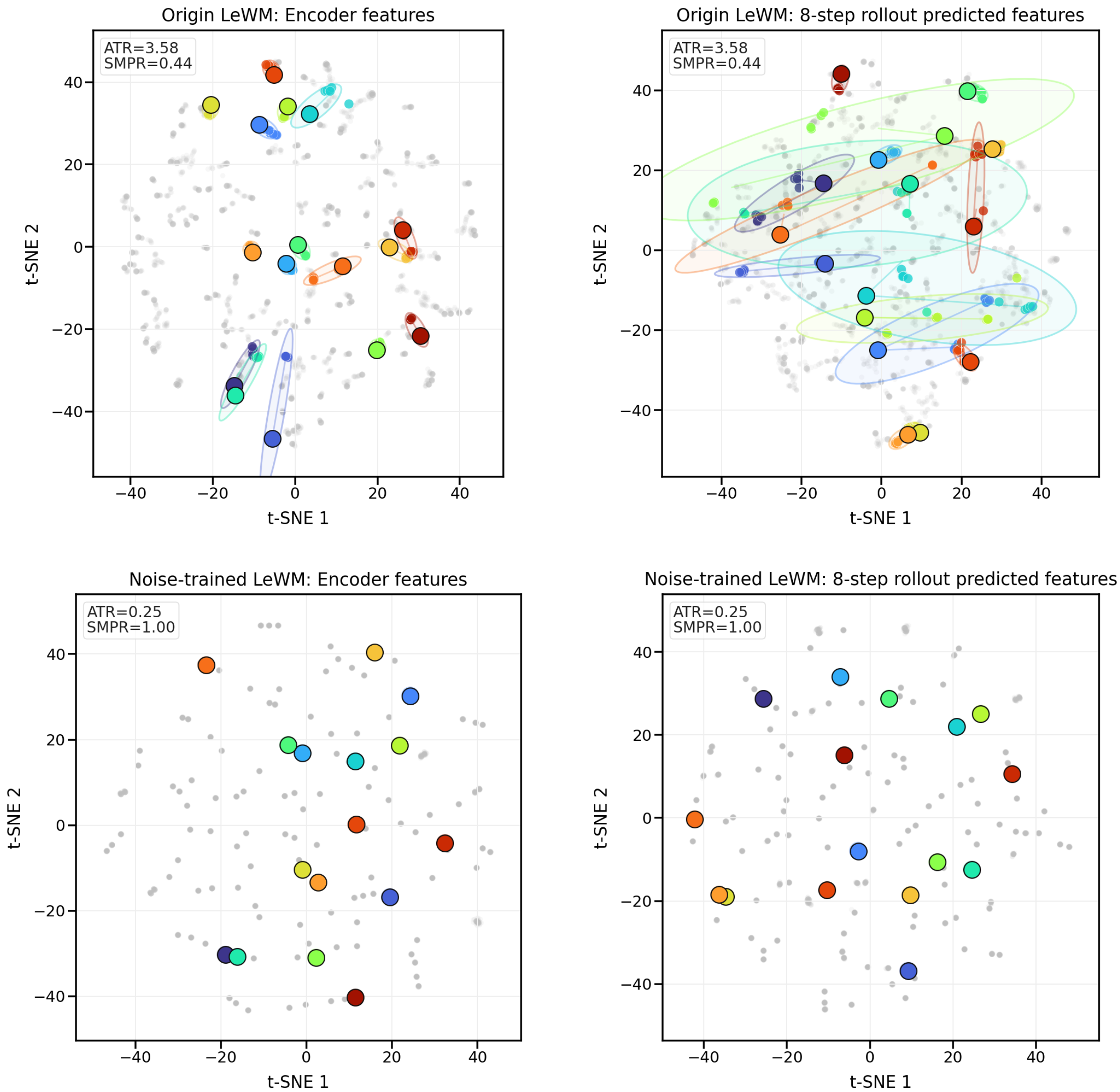

Figure 7: The PushT t-SNE view provides qualitative intuition only and is not used for quantitative claims. The four panels compare LeWM-base with the $\sigma_{\max} = 0.08$ noise-trained checkpoint in encoder features and 8-step rollout predicted features for repeated same-state perturbation views; colored clusters mark fixed random anchors. ATR and SMPR are computed in the original projected-rollout diagnostic space.

E Fixed-pool Planner Audit Details

E.1 Radius–margin parameter interpretation

Theorem 1 separates the quantities needed for a numerical fixed-pool probability bound from the lower-sample diagnostic summaries reported in the experiments. In the fixed-pool diagnostic summaries used for Figures 3 and 9, the candidate count is $K = 65$ for every task–checkpoint row. The theorem's single-candidate tail probability α is not fixed by the q90 reporting level used by ATR. A q90 summary descriptively leaves 10% of measured same-state radii above the reported quantile, but that descriptive exceedance rate is not the α

used in the theorem. If it were naively inserted into the union-bound term, the result would be $65 \times 0.1 = 6.5$, which is vacuous. A non-vacuous numerical candidate-flip bound would require a much smaller single-candidate tail probability or a different concentration argument. We therefore use q90 ATR and the available q50/q90 cost-margin overlap as diagnostic evidence for the radius–margin mechanism, not as a calibrated probability guarantee.

Table 15: Fixed-pool radius–margin interpretation used in the diagnostic analysis. The behavioral recovery band is defined from closed-loop observation-only $\sigma = 0.08$ scores; the proxy-positive range uses $\widehat{Q}_{0.50}(\Delta_{\mathcal{A}}) - 2\widehat{Q}_{0.90}(|C_h - C_{\tilde{h}}|) > 0$ on the available summary-level fixed-pool summaries. These ranges explain the recovery/proxy bands in Figure 3 and the overlap proxy in Figure 9; they are not used to claim a calibrated flip-probability bound.

Task	K	recovery band	proxy-positive range	interpretation
TwoRoom	65	0.01–0.08	0.02–0.08	proxy misses the first recovery-band checkpoint
PushT	65	0.03–0.08	0.02–0.08	proxy includes one earlier checkpoint
Reacher	65	0.02–0.08	0.04–0.08	proxy is conservative at low noise-training levels
Cube	65	0.03–0.08	0.03–0.08	proxy and recovery-band ranges align

The one-grid tolerance is $\tau_{\text{grid}} = 0.01$ in $\sigma_{\max}$ units. Recovery-onset error is defined as $\rho^{\text{diag}}_{\text{start}} - \rho^{\text{beh}}_{\text{start}}$, so positive values mean the diagnostic proxy is later and negative values mean it fires earlier.

Table 16: Boundary-aware interpretation of the fixed-pool radius–margin proxy. Equal spacing in $\sigma_{\max}$ is not equal spacing in task sensitivity, so these rows are interpreted as task-conditioned boundary comparisons rather than grid-point classifier scores.

Task	recovery band	proxy range	recovery-onset error	within τ_{grid}?	interpretation
TwoRoom	0.01–0.08	0.02–0.08	+0.01	yes	proxy misses the first recovery-band checkpoint
PushT	0.03–0.08	0.02–0.08	−0.01	yes	proxy fires one grid step early
Reacher	0.02–0.08	0.04–0.08	+0.02	partial	fixed-pool cost-margin proxy is conservative
Cube	0.03–0.08	0.03–0.08	0.00	yes	proxy and recovery band align

The measured fixed-pool flip rate is the clean/noisy top-1 disagreement rate under the same candidate set, so one minus this flip rate gives an empirical top-1 agreement analysis that is closer to the theorem's intermediate statement than closed-loop score. It still does not cover adaptive CEM resampling or repeated replanning. Figure 8 shows the full-sweep planner-side traces that were intentionally separated from the main full-sweep ACPC figure.

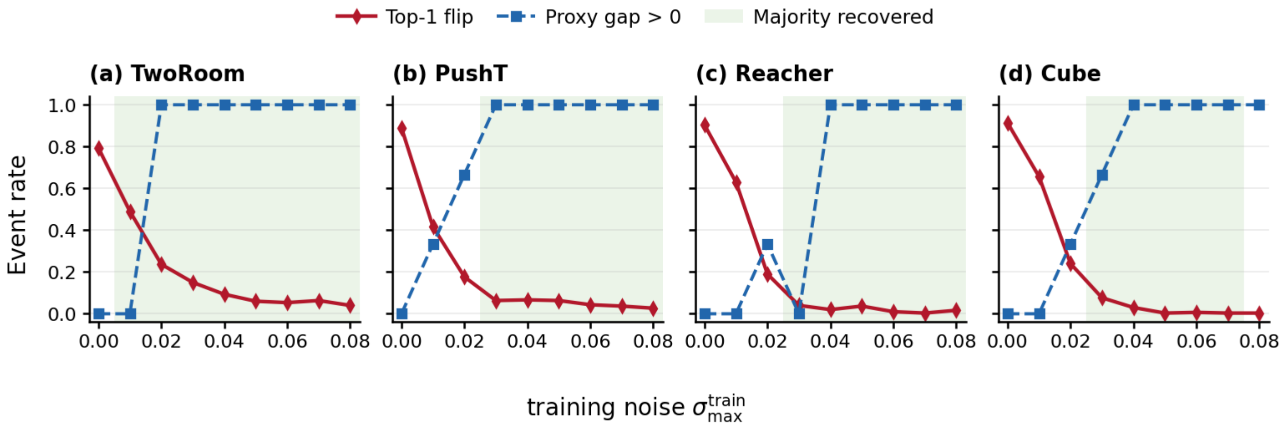

Figure 8: Planner-side fixed-pool traces are reported separately from the main ACPC full-sweep figure. Continuous green bands mark training-noise ranges where a majority of seed rows meet the behavioral recovery criterion; top-1 flip is observed clean/noisy fixed-pool top-candidate disagreement, and proxy gap > 0 marks rows where the retained q50/q90 summary-level radius–margin proxy is positive. These traces support the planner-side mechanism audit but are not a checkpoint-selection rule.

Table 17: Fixed-pool top-1 agreement analysis derived from the measured fixed-pool flip rate. Rows show the no-noise baseline, the first closed-loop recovery-band checkpoint on the discrete sweep, and the matched $\sigma_{\max} = 0.08$ endpoint.

Task	checkpoint	cost drift q90	clean margin q50	proxy gap	top-1 agreement
TwoRoom	base 0.00	180.33	94.14	−266.52	0.207
TwoRoom	onset 0.01	85.42	100.56	−70.28	0.510
TwoRoom	endpoint 0.08	6.00	65.46	53.46	0.960
PushT	base 0.00	163.82	129.21	−198.44	0.110
PushT	onset 0.03	18.16	143.29	106.96	0.937
PushT	endpoint 0.08	7.15	132.05	117.74	0.973
Reacher	base 0.00	217.30	17.29	−417.32	0.097
Reacher	onset 0.02	37.67	18.54	−56.80	0.810
Reacher	endpoint 0.08	4.24	20.08	11.59	0.983
Cube	base 0.00	198.70	82.76	−314.65	0.087
Cube	onset 0.03	39.48	82.95	3.99	0.923
Cube	endpoint 0.08	5.05	76.94	66.85	0.997

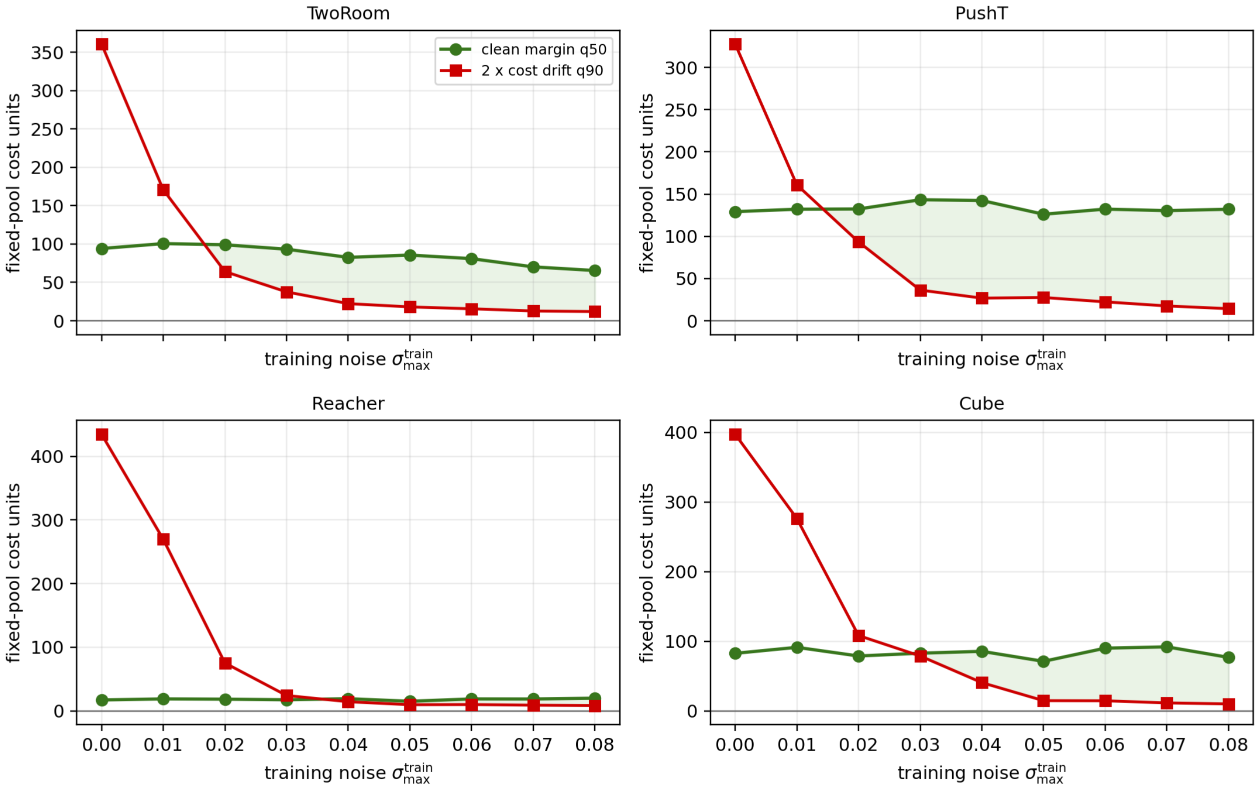

Figure 9: Fixed-pool radius–margin overlap proxy across the full Gaussian training sweep. Curves use the available summary-level fixed-pool summaries: q50 clean candidate margin and twice the q90 paired cost drift. The separate sample-level recomputation in Table 20 provides q10/q95 gaps and event rates; q10/q95 gaps remain negative, so the analysis supports the mechanism rather than a calibrated planner-margin bound. The sampled pool is fixed rather than adaptive CEM.

E.2 Fixed-pool event-rate audit

For each sampled state h_i, let $E_{\text{pool}}(h_i)$ be the fixed-pool sufficient event that the maximum paired candidate-cost drift over the $K = 65$ fixed candidates is below one half of the clean top-1/top-2 margin. Let $F_{\text{pool}}(h_i)$ be the observed clean/noisy fixed-pool top-1 disagreement. We report three empirical rates over the sampled anchors in each split:

$$\hat{p}_{\text{cert}} = \frac{1}{n}\sum_i \mathbf{1}[E_{\text{pool}}(h_i)],$$
$$\hat{p}_{\text{flip}} = \frac{1}{n}\sum_i \mathbf{1}[F_{\text{pool}}(h_i)],$$
$$\hat{p}_{\text{flip|cert}} = \frac{\sum_i \mathbf{1}[F_{\text{pool}}(h_i) \wedge E_{\text{pool}}(h_i)]}{\sum_i \mathbf{1}[E_{\text{pool}}(h_i)]}. \tag{14}$$

The sufficient-event interpretation is sample-level: whenever $E_{\text{pool}}(h_i)$ holds, the corresponding deterministic condition rules out a flip for that history and candidate set. Consequently $\hat{p}_{\text{flip|cert}} = 0$ is an implementation invariant, not an independent empirical validation statistic. The older coarse rates below are retained for provenance; the main result uses the sharper candidate-wise slack, risk–coverage curve, and block bootstrap.

Table 18: Sharp fixed-pool certificate coverage and observed risk by task–training-seed block.

Block	Coarse cov.	Sharp cov.	Flip rate	Flip | sharp fail
Cube–3072	0.578	0.751	0.183	0.737
Cube–3073	0.461	0.682	0.211	0.664
Cube–3074	0.461	0.637	0.293	0.807
PushT–3072	0.462	0.664	0.204	0.609
PushT–3073	0.508	0.661	0.224	0.662
PushT–3074	0.514	0.698	0.202	0.669
Reacher–3072	0.442	0.760	0.176	0.731
Reacher–3073	0.360	0.687	0.210	0.670
Reacher–3074	0.344	0.663	0.224	0.667
TwoRoom–3072	0.518	0.724	0.189	0.685
TwoRoom–3073	0.556	0.757	0.173	0.712
TwoRoom–3074	0.526	0.720	0.203	0.726

In this interpretation, β_{plan} and β_{disc} name empirical failure components rather than calibrated probabilities. The planning component is represented by the fixed-pool cost-margin proxy, top-1 agreement analysis, and the sample-level event-rate recomputation; the discriminability component is represented directly by SMPR, where higher values indicate stronger margin preservation. The closed-loop recovery band explains the fixed-pool radius–margin validation rather than converting the separate ATR/SMPR diagnostic-region validation into a threshold-classifier claim.

Table 19: Summary-level fixed-pool top-1 agreement rises as the q50/q90 proxy gap turns positive. Agreement is one minus the measured fixed-pool flip rate; cert-pass rates require sample-level maximum cost-drift traces.

Task	row	$\sigma_{\max}^{\text{train}}$	top-1 agreement	proxy gap	proxy pass rate
TwoRoom	base	0.00	0.207	−266.52	0.00
TwoRoom	proxy onset	0.02	0.763	34.93	1.00
TwoRoom	endpoint	0.08	0.960	53.46	1.00
PushT	base	0.00	0.110	−198.44	0.00
PushT	proxy onset	0.02	0.823	38.95	0.67
PushT	endpoint	0.08	0.973	117.74	1.00
Reacher	base	0.00	0.097	−417.32	0.00
Reacher	proxy onset	0.04	0.980	4.79	1.00
Reacher	endpoint	0.08	0.983	11.59	1.00
Cube	base	0.00	0.087	−314.65	0.00
Cube	proxy onset	0.03	0.923	3.99	0.67
Cube	endpoint	0.08	0.997	66.85	1.00

Table 20: Full-sweep sample-level fixed-pool event-rate calibration. Rows split all task–seed–$\sigma_{\max}$ checkpoints by the closed-loop recovery-band label. Cert-pass is the fraction of sampled states satisfying the fixed-pool sufficient event; top-1 flip is the observed clean/noisy fixed-pool best-candidate disagreement. These event rates strengthen the mechanism analysis but are not calibrated probability bounds.

Task	cert-pass fragile → recovered	top-1 flip fragile → recovered	q10/q95 gap fragile → recovered
TwoRoom	0.05 → 0.67	0.79 → 0.07	-1053.7 → -269.0
PushT	0.16 → 0.67	0.38 → 0.06	-797.6 → -193.5
Reacher	0.06 → 0.54	0.54 → 0.02	-873.0 → -29.3
Cube	0.08 → 0.78	0.53 → 0.01	-713.4 → -55.3
ALL	0.06 → 0.61	0.54 → 0.04	-847.7 → -136.6

Table 21: Finite-sample empirical fixed-pool risk audit. Each cell reports fragile → recovered rate [95% Wilson interval] over sampled fixed-pool anchors. The conditional column restricts the denominator to cert-pass anchors, directly evaluating whether the sufficient event is associated with low fixed-pool candidate-flip risk. These intervals quantify event-rate estimation uncertainty and are not calibrated theorem probability bounds.

Task	cert-pass	top-1 flip	flip | cert-pass
TwoRoom	0.09 [0.06,0.12] → 0.61 [0.59,0.63]	0.72 [0.68,0.77] → 0.10 [0.08,0.11]	0.00 [0.00,0.10] → 0.00 [0.00,0.00]
PushT	0.18 [0.16,0.21] → 0.65 [0.63,0.67]	0.52 [0.49,0.55] → 0.06 [0.05,0.07]	0.00 [0.00,0.02] → 0.00 [0.00,0.00]
Reacher	0.11 [0.09,0.13] → 0.52 [0.50,0.54]	0.55 [0.51,0.58] → 0.03 [0.02,0.04]	0.00 [0.00,0.04] → 0.00 [0.00,0.00]
Cube	0.32 [0.29,0.35] → 0.64 [0.62,0.67]	0.47 [0.44,0.50] → 0.04 [0.03,0.05]	0.00 [0.00,0.01] → 0.00 [0.00,0.00]
ALL	0.20 [0.19,0.21] → 0.61 [0.59,0.62]	0.53 [0.52,0.55] → 0.06 [0.05,0.06]	0.00 [0.00,0.01] → 0.00 [0.00,0.00]

Table 22: Sample-level fixed-pool endpoint calibration. Cert-pass is the fraction of sampled states where the maximum paired candidate-cost drift over the fixed 65-candidate pool is below half the clean top-1/top-2 margin. Values are mean $\pm$ population std across training seeds 3072/3073/3074 with 100 sampled states per seed. Strict q10/q95 gaps remain negative, so these rows support a fixed-pool mechanism analysis rather than a calibrated probability certificate.

Task	cert-pass base $\to \sigma_{\max} = 0.08$	top-1 flip base $\to \sigma_{\max} = 0.08$	q10/q95 gap base $\to \sigma_{\max} = 0.08$
TwoRoom	$0.05 \pm 0.00 \to 0.73 \pm 0.03$	$0.80 \pm 0.02 \to 0.03 \pm 0.00$	$-1084.6 \pm 44.0 \to -179.0 \pm 44.2$
PushT	$0.01 \pm 0.01 \to 0.74 \pm 0.04$	$0.92 \pm 0.03 \to 0.03 \pm 0.02$	$-917.4 \pm 44.3 \to -126.2 \pm 13.5$
Reacher	$0.00 \pm 0.00 \to 0.61 \pm 0.01$	$0.88 \pm 0.03 \to 0.01 \pm 0.00$	$-935.2 \pm 32.1 \to -24.7 \pm 2.3$
Cube	$0.00 \pm 0.00 \to 0.83 \pm 0.01$	$0.92 \pm 0.03 \to 0.00 \pm 0.00$	$-1005.3 \pm 13.9 \to -31.3 \pm 7.7$

F Local Sensitivity Details

The main matched-map calibration in Figure 5 and table 5 compares finite differences and covariance-aware JVPs on the same map and anchors. This appendix separates the raw encoder, rollout, and composed JVP/Hutchinson trace ratios from the relative isotropic gain; the latter can exceed one and is not an angle or certificate.

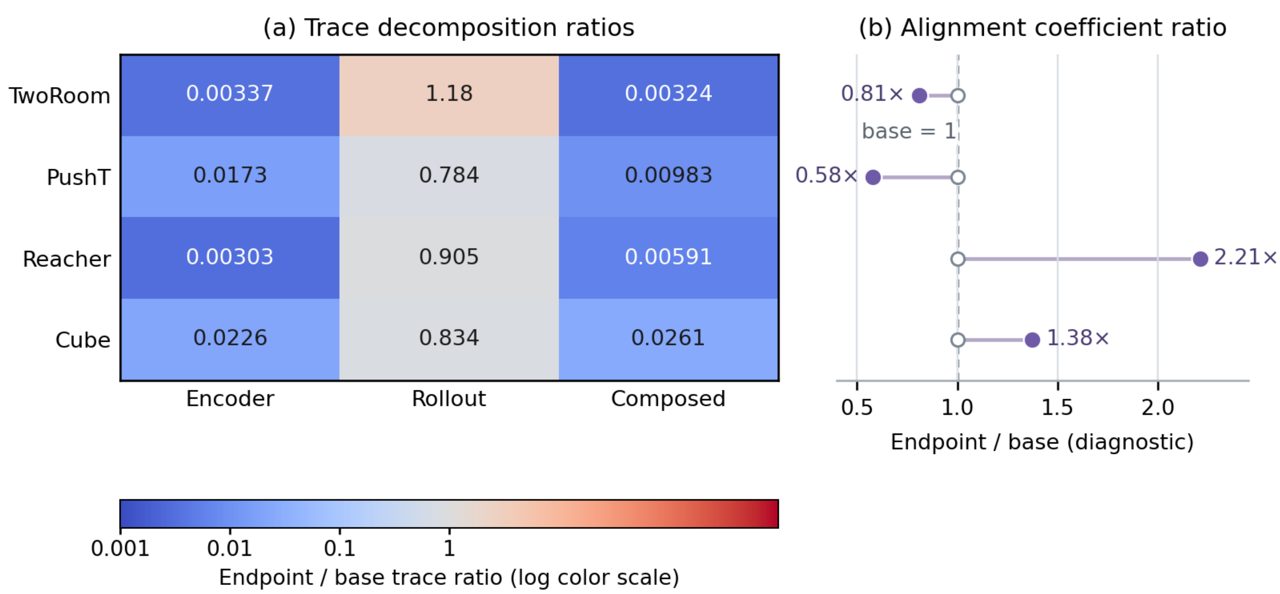

Figure 10: JVP/Hutchinson decomposition behind the local sensitivity audit. Raw trace ratios show encoder, rollout, and composed endpoint/base sensitivity estimates; the relative isotropic gain is displayed separately because it is not on the same scale as the trace quantities. These local diagnostics explain where composed sensitivity changes, but they do not provide a global or closed-loop guarantee.

Table 23: Exact-JVP/Hutchinson local sensitivity decomposition. Every entry is an endpoint/base ratio, using 16 sampled sequences and 8 Rademacher probes per checkpoint, then taking medians over training seeds when multiple seeds are present. The first three columns are ratios of the raw encoder, rollout, and composed Frobenius-trace estimates; the constant per-input-dimension normalizers cancel within each endpoint/base comparison. The final column reports `kappa_relative_isotropic`, the relative isotropic alignment gain $\kappa_{\mathrm{rel}} = d_z \|J_G J_E\|_F^2 / (\|J_G\|_F^2 \|J_E\|_F^2 + \varepsilon)$. This diagnostic can exceed 1 and is not an angle, cosine, or certificate. Exact autograd JVPs with math/eager attention estimate local traces, not a full Jacobian matrix or a closed-loop guarantee.

Task	raw encoder trace	raw rollout trace	raw composed trace	relative isotropic gain
TwoRoom	0.004	1.256	0.003	0.644
PushT	0.017	0.995	0.010	0.633
Reacher	0.003	0.887	0.006	1.739
Cube	0.023	1.079	0.028	1.042

G Cross-Stressor Scope

The main text reports every fixed-severity blur/resize base-to-endpoint pair used by the paired-change analysis, not only favorable task aggregates. The supported result is one relative zero-threshold rule within the calibrated LeWM family. Additional corruption families, severities, and model-family replications remain outside this release, so the result is not a universal corruption or cross-model calibration theorem.

H Reproducibility

The release checks are deterministic over the retained artifacts. Summary-level diagnostics and checkpoint-level audits are rebuilt by repository commands documented outside the paper; checkpoint-level audits require accessible LeWM checkpoints and datasets. The PDF and source package are checked by the repository readiness target, and the author-placeholder override is for local readiness checks only.